A natural boundary for the low-temperature bulk square-lattice Ising susceptibility

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Abstract

The unit circle is a natural boundary for the normalized zero-field susceptibility of the isotropic infinite square lattice, continued from the low-temperature ordered pure phase. At a dense cyclotomic family, an average-angle residue followed by a convergent shape integral gives a nonzero singular derivative of the first resonant form factor. A weighted contour deformation, parameter disks, and uniform sector estimates bound the complete higher-particle tail at every order required by the product rule. The identification with the connected bulk response uses the published correlation expansion and spontaneous magnetization, with the onsite term, magnetization subtraction, and full even-particle sum. We evaluate the leading amplitude, compare the four angular contributions with the formula in the literature, and give a finite-order obstruction to smooth extension on every open boundary arc.

1 Introduction

Let $s = \sinh(2\beta_{\mathrm{phys}}J)$, with inverse temperature β_{phys}, zero magnetic field, and equal positive nearest-neighbor coupling J on the infinite square lattice. Write

$$\mathcal{X}(s) = \beta_{\mathrm{phys}}^{-1}\chi(s), \qquad \mathcal{M}^2(s) = (1 - s^{-4})^{1/4},$$

The factor $\mathcal{M}^2$ is the square of the low-temperature spontaneous magnetization [13]; its branch tends to one at infinity. The symbol β_{phys} distinguishes inverse temperature from the geometric angle β used below. We study the exterior germ inherited from real $s > 1$. On that real interval the response is defined by connected correlations in Palmer–Tracy's infinite-volume + state, obtained as the limit of + boundary conditions [8, pp. 332, 361]. Global spin flip gives the same two-point function and the same connected correlation after subtracting $\mathcal{M}^2$ in the − state. The full bulk response $\mathcal{X}$ includes the onsite term. The natural-boundary theorem states that

no point of $|s| = 1$ admits a local holomorphic continuation of this exterior germ.

The physical susceptibility uses the local branches of $\beta_{\mathrm{phys}}(s) = \operatorname{arsinh}(s)/(2J)$ continued from the positive real interval. These branches are analytic and nonzero at each selected nonaxis point. Thus the normalized and physical responses have the same local obstruction.

For an exterior function, a dense set of points where a fixed finite derivative at each point is unbounded along an exterior radius suffices to show that $\mathbb{T}$ is a natural boundary.

1.1 Related work and the bulk problem

The form-factor expansion of the bulk susceptibility goes back to Wu, McCoy, Tracy, and Barouch [12]. Later correlation representations include the low-temperature Fredholm formula of Palmer and Tracy [8], used in Appendix 1 of Tracy and Widom [10]. Guttmann and Enting [3] supplied early numerical evidence for a natural boundary. Nickel [5, 6] identified the unit-circle form-factor singularities now bearing his name; Boukraa, Hassani, Maillard, and Zenine [1] compared Landau conditions, ODE singularities, and actual singularities of holonomic Ising-class integrals, including susceptibility-related singularity analysis. Boukraa et al. [2] developed further series and singularity evidence. Orrick, Nickel, Guttmann, and Perk [7, Section 3.2] examined the dependence of the singularity amplitudes on particle order as evidence against cancellation in the bulk sum.

For the diagonal susceptibility, Tracy and Widom [9] proved a natural-boundary theorem. Their Section IV isolates a divergent derivative of one particle term and bounds the corresponding derivative of the remaining sum. Their later work [11] extends that direction to more general sums of Toeplitz determinants. The analysis in Sections III–IV of [11] uses contour partitions, Vandermonde homogeneity, and factorial decay from Hadamard bounds. For the bulk susceptibility the two lattice directions give the global factors $(1-X)(1-Y)$ in (2.1). The local amplitude and tail estimates below account for both factors. We also use the gradient method of Tracy and Widom's fixed-order analysis [10] to obtain the radial bounds in Section 4.

1.2 Leading derivative and tail estimates

For each selected point s_*, Section 3 specifies a first even order $N_0 = 2p$ and the fixed integer $k = N_0^2/2 - 1$. Put $s_\varepsilon = (1+\varepsilon)s_*$. The two estimates are

$$T_{N_0}^{(k)}(s_\varepsilon) = 2L_\beta \varepsilon^{-1/2} + o(\varepsilon^{-1/2}), \qquad L_\beta \neq 0, \tag{1.1}$$

$$\sum_{\substack{N>N_0 \\ N \text{ even}}} |T_N^{(j)}(s_\varepsilon)| = o(\varepsilon^{-1/2}), \qquad 0 \le j \le k. \tag{1.2}$$

The finitely many lower form factors and the lower derivatives of the first singular form factor remain bounded. The normalization in Section 2 then gives

$$\mathcal{X}^{(k)}(s_\varepsilon) = 4\mathcal{M}^2(s_*) L_\beta \varepsilon^{-1/2} + o(\varepsilon^{-1/2}).$$

Leibniz's rule uses the full derivative range in (1.2), since derivatives may fall on the magnetization factor. The proof of (1.1) uses the arithmetic in Section 3 and the residue in Section 4. For the tail, the identity $T_N = F_N + K_N + S_N$ in Section 5 includes the weighted Stokes correction S_N. Sections 6–8 estimate these integrals using the fixed-domain differentiation rules in Appendix D. Appendix E gives the order of constant choices and the coverage of the tail sectors.

The boundary point and differentiation order are fixed throughout. Uniformity in N refers to the indicated particle window. The contour radius is held fixed during differentiation, and the windows depending on ε are introduced in the resulting estimates.

Symbol	Meaning
N, N_0, k	Even particle number; first singular order; fixed derivative order.
S, W, z, ϕ	Dispersion variables, with $z = e^{-\mathrm{i}\phi}$ on the lower sheet.
Y, Z, P_{ij}	Products of the y, z coordinates and the complete canceled pair.
p, m, a, P	Fixed deformation/selector functions and occupancy $P = \sum_i p_i$.
H, λ_*	$\log(1/\varepsilon)$ and the post-differentiation current split ε^{α_0}.
b_N, ρ_N	Exponential microcore and full-equality scales in the intermediate window.

AI Declaration

ChatGPT was used extensively in the mathematical research, proof development, and preparation of this manuscript.

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2 The normalized bulk representation

This section fixes the observable and its integration measures before any asymptotic argument. The offsite form factors below reconstruct the full connected susceptibility through the onsite term and the magnetization prefactor. The residue reduction then gives the one-contour form used for the tail estimates. Put $S(s) = s + s^{-1}$ and

$$D(x, y; s) = S(s) - \tfrac{1}{2}(x + x^{-1}) - \tfrac{1}{2}(y + y^{-1}).$$

We take as external physical input the low-temperature correlation expansion in the published Appendix 1, equations (7)–(8), pp. 1132–1133 of [10]. Its upstream source is [8, Cor. 2.0, p. 351; Thm. 4.1, p. 361; Thm. 5.0, p. 374, and proof pp. 375–378]. For even N, let T_N denote the Fredholm-normalized offsite form factor. With positively oriented circles sufficiently close to the unit circle, it is

$$T_N(s) = \frac{1}{N!} \oint \cdots \oint \frac{X^{-1} + Y^{-1}}{(1 - X)(1 - Y)} \prod_{i<j} \frac{x_i - x_j}{1 - x_i x_j} \frac{y_i - y_j}{1 - y_i y_j} \prod_{i=1}^{N} \frac{\mathrm{d}x_i \, \mathrm{d}y_i}{(2\pi\mathrm{i})^2 D(x_i, y_i; s)}, \tag{2.1}$$

$$X = \prod_i x_i, \qquad Y = \prod_i y_i,$$

$$\mathcal{X}(s) = 1 - \mathcal{M}^2(s) + 2\mathcal{M}^2(s) \sum_{\substack{N \geq 2 \\ N \text{ even}}} T_N(s). \tag{2.2}$$

To obtain this normalization, convert the N normalized angular measures in Appendix 1, equation (7), of [10] and retain the N normalized auxiliary residues. It agrees with the prefactor $(2\pi\mathrm{i})^{-2N}/N!$ displayed with the published equation (3). That coefficient sums all sites, including the origin, and equals $f_{00}^{(N)} + 2T_N$ at fixed even N. Appendix F defines these terms and derives the relation.

Lemma 2.1 (Residue reduction)**.** *Let*

$$W(y; s) = S(s) - \tfrac{1}{2}(y + y^{-1}), \quad z = W - \sqrt{W^2 - 1} = e^{-\gamma}, \quad R(y; s) = \frac{2z^2}{1 - z^2}.$$

Choose the square root by continuation of the interior root and require $|z(y; s)| < r < 1$ on the y-circle. Then

$$T_N = \frac{1}{N!} \oint \cdots \oint \frac{Z^{-1} + Y^{-1}}{(1 - Z)(1 - Y)} \prod_{i<j} P_{ij} \prod_i R(y_i; s) \frac{\mathrm{d}y_i}{2\pi\mathrm{i}}, \qquad Z = \prod_i z_i, \tag{2.3}$$

where

$$P_{ij} = \frac{z_i - z_j}{1 - z_i z_j} \frac{y_i - y_j}{1 - y_i y_j} = -\frac{(y_i - y_j)^2 z_i z_j}{y_i y_j (1 - z_i z_j)^2}. \tag{2.4}$$

Proof. The only enclosed dispersion root in each x_i is z_i; its residue is $2z_i^2/(1-z_i^2)$. At $x_i = 0$ the reciprocal dispersion factor is $-2x_i + O(x_i^2)$, canceling the apparent numerator pole. All pair and product poles involving the other interior x's are outside the x_i-circle. The same pole separation holds at every subsequent residue: each root already substituted has modulus less than r. Subtracting $z_i + z_i^{-1} = 2S - y_i - y_i^{-1}$ for two indices gives

$$(z_i - z_j)(1 - z_i z_j) = -\frac{z_i z_j}{y_i y_j}(y_i - y_j)(1 - y_i y_j),$$

which proves the canceled expression. Each residue consumes its own normalized x-measure and leaves a normalized y-measure. □

For a fixed $s_* = e^{\mathrm{i}\theta_*}$, $0 < \theta_* < \pi/2$, set $s_\varepsilon = (1+\varepsilon)s_*$. On a disk $|s - s_\varepsilon| < c_1\varepsilon$ choose $r = e^{-c_0\varepsilon}$, with $c_0, c_1 > 0$ sufficiently small. Then $\Im W \geq c_2\varepsilon$, uniformly in the angle. The inverse root is in the lower unit half-disk. Writing $\gamma = u + \mathrm{i}v$ gives $\Im W = \sinh u \sin v \leq \sinh u$, hence $u \geq c_3\varepsilon$. Choosing $c_0 < c_3$ gives $|z| < r$, placing the root inside the residue contour. During an s-derivative this chosen r is held fixed.

Lemma 2.2 (Symmetries)**.** *The exterior normalized susceptibility and its even form factors are even in s and invariant under conjugation of the argument and value.*

Proof. Both symmetries are checked on the normalized contour representation, so they apply to the complete bulk function as well as to each even form factor.

In (2.1) replace s, x_i, y_i by $-s, -x_i, -y_i$. Each dispersion factor changes sign; their number is even. Each pair changes sign in both variables and each product is unchanged. The contour substitution preserves the integral. The branch of $\mathcal{M}^2$ is also even. Conjugation follows either by conjugating these contours with their orientations and measures together or from the real low-temperature Fredholm expansion and the identity theorem. □

3 A dense family with a unique first cosine pair

The prime family has a unique resonant cosine pair at its first even order. This gives two ordered local charts. Nonresonance at the dispersion endpoint supplies exponential separation for the all-branch tail. For this paper a unit-circle point is a Nickel point of order n when

$$s_* + s_*^{-1} = \cos(2\pi l/n) + \cos(2\pi m/n) \quad \text{for some integers } l, m.$$

This is Nickel's singularity condition [5, 6], in the convention of [10]. Only even orders occur in the low-temperature expansion under study. The theorem below establishes the additional first-order and uniqueness properties for the specific point family used here.

Theorem 3.1. *Let $p \geq 11$ be prime, and choose distinct integers a, b with $\alpha = 2\pi a/p$, $\beta = 2\pi b/p \in (0, \pi/2)$. Set*

$$c = \frac{\cos\alpha + \cos\beta}{2}, \qquad s_* = e^{\mathrm{i}\arccos c}, \qquad N_0 = 2p.$$

The first even Nickel order at s_* is N_0. At that order the unordered cosine pair is uniquely $\{\cos\alpha, \cos\beta\}$. These points are dense in the open upper-right arc. If $\theta_B = \arccos(2c-1)$, then $e^{-\mathrm{i}\theta_B}$ is algebraic and is not a root of unity.

Proof. The proof has four parts: exclude smaller even orders, establish uniqueness at the first order, prove density, and rule out a root of unity at the branch endpoint.

Let $\zeta = e^{2\pi \mathrm{i}/p}$, $d = (p-1)/2$, $u_j = \zeta^j + \zeta^{-j}$, and $E = \mathbb{Q}(\zeta + \zeta^{-1})$. The trace of every nonzero-index u_j is -1, by summing over the pairs of nonzero residue classes. Thus $\mathrm{Tr}_E(2c) = -1$. If $2c$ were rational it would be $-1/d$, whereas it is positive.

First even order. If an even $n < 2p$ represented the same Nickel point, then $p \nmid n$ and $c \in \mathbb{Q}(\zeta_p) \cap \mathbb{Q}(\zeta_n) = \mathbb{Q}$; the last equality is proved in Lemma A.1. This contradicts the preceding trace. Order $2p$ contains the indicated pair of angles and is therefore first.

Uniqueness. At order $2p$ every cosine is $\pm u_j/2$ or ± 1. The only rational relation between $1, u_1, \dots, u_d$ is a multiple of $1 + \sum_j u_j = 0$: lift a relation to a polynomial of degree at most $p-1$ vanishing at ζ, and use its minimal polynomial $1 + X + \cdots + X^{p-1}$. An equality between two cosine pairs uses at most four of the u_j. Since $d \geq 5$, one coefficient is zero, forcing that multiple to vanish. The remaining coefficients give precisely the two positive terms u_a, u_b, with no constant term.

Density. For any $c_0 \in (0,1)$, along unbounded primes choose two distinct neighboring grid angles tending to $\arccos c_0$. They eventually lie in $(0, \pi/2)$, and their cosine averages tend to c_0.

Branch nonresonance. Finally $w = 2\cos\theta_B = u_a + u_b - 2$ has trace $-2d-2$. Were $e^{\mathrm{i}\theta_B}$ a root of unity, every conjugate of w would be in $[-2,2]$; every embedding of E restricts to such a conjugate. Its trace could not be less than $-2d$. Algebraicity follows from $x^2 - wx + 1 = 0$. □

Lemma 3.2 (Exponential separation)**.** *For the fixed algebraic number $\xi = e^{-\mathrm{i}\theta_B}$ there is $A > 0$ such that $|1 - \xi^N| \geq e^{-AN}$ for every $N \geq 1$.*

Proof. A nonzero integer resultant has absolute value at least one. Bounding all conjugate factors except the selected one converts this arithmetic fact into the required exponential lower bound.

Write its primitive integer minimal polynomial as $P(X) = a_d \prod_{\nu=1}^{d} (X - \xi_\nu)$, with $\xi_1 = \xi$. It has no root in common with $X^N - 1$: such a conjugate root would make its irreducible polynomial cyclotomic and would make ξ a root of unity. Consequently

$$1 \leq |\operatorname{Res}(P, X^N - 1)| = |a_d|^N \prod_{\nu} |\xi_\nu^N - 1| \leq |a_d|^N |1 - \xi^N|\, 2^{d-1} \prod_{\nu > 1} \max(1, |\xi_\nu|)^N.$$

Increasing A absorbs all fixed factors for $N \geq 1$. □

4 The first singular form factor

We localize the double integral by the unreduced angles and take the mean residue before rescaling the shape variables. The complement estimate then gives the asymptotic for the complete form factor. Fix the family of Theorem 3.1 and put

$$N = N_0, \qquad k = N^2/2 - 1, \qquad S_* = \cos\alpha + \cos\beta = 2\cos\theta_*.$$

All constants in this section are allowed to depend on this fixed N. At the ordered lower chart $y_0 = e^{-\mathrm{i}\alpha}$, $z_0 = e^{-\mathrm{i}\beta}$, write $y_j = re^{\mathrm{i}(-\alpha + u_j)}$ and $z_j = e^{-\mathrm{i}\phi_j}$. Define

$$v = \sum_j u_j, \qquad t_j = u_j - v/N, \qquad \sum_j t_j = 0.$$

The absolute Jacobian of $(v, t_1, \dots, t_{N-1}) \mapsto (u_1, \dots, u_N)$ is one.

Compatible localization. Let $\Omega^{\mathrm{dbl}}_{N,s}$ denote the complete normalized double-contour density in (2.1). Around each of the two all-equal lower y-configurations, choose a smooth angular weight

$$w_\gamma(\theta) = \eta_\gamma(v_\gamma)\chi_\gamma(t_\gamma), \qquad \gamma \in \{\alpha, \beta\}.$$

Here $u_{\gamma,j} = \theta_j + \gamma$, $v_\gamma = \sum_j u_{\gamma,j}$, and $t_{\gamma,j} = u_{\gamma,j} - v_\gamma/N$. Choose $\eta_\gamma = 1$ on $|v_\gamma| \le \delta/2$, supported in $|v_\gamma| < \delta$, and $\chi_\gamma = 1$ near the zero shape, supported in a fixed small shape ball. Extend each weight smoothly by zero on the angular torus, with disjoint supports and values in $[0, 1]$. The weights depend only on the y angles and are fixed during the parameter differentiation and the x and mean residues. This gives the exact identity

$$T_N = I_\alpha + I_\beta + I_{\mathrm{rem}}, \qquad I_\gamma = \int w_\gamma \Omega^{\mathrm{dbl}}_{N,s}, \qquad I_{\mathrm{rem}} = \int (1 - w_\alpha - w_\beta)\Omega^{\mathrm{dbl}}_{N,s}$$

holds before any localization in x. The remaining weight vanishes on an open neighborhood of both bad double-torus configurations. The classification and estimate in Lemma 4.3 therefore give $\partial_s^j I_{\mathrm{rem}} = O(1)$ for every fixed j. Its further smooth partition may depend on both sets of angles, since no residue calculation is applied to these complementary pieces.

Because every x integration in I_γ remains unweighted, Lemma 2.1 applies directly and leaves exactly the same y-weight on the reduced integral. In this reduced local integral replace $\eta_\gamma(v)$ by the indicator of $[-\delta, \delta]$. Their difference is supported where $\delta/2 \le |v| \le \delta$. Using the coefficients b, d computed below, choose the shape radius h so that $dh^2 \ll b\delta$, and then take δ and the parameter neighborhood sufficiently small. On this edge region, the Y phase is separated by $|v|$, while the distance of $\Re \sum_j \phi_j$ from $2\pi\mathbb{Z}$ is bounded below by $b|v| - C(h^2 + v^2 + \varepsilon)$. Both global denominators, and every local one-body and pair denominator, are therefore uniformly separated from zero. The replacement changes each fixed s-derivative by $O(1)$, with the radius held fixed during differentiation. Only this hard mean segment is moved in Lemma 4.1; the shape cutoff stays fixed. No nonholomorphic angular weight is moved through an x residue or a complex mean deformation.

Lemma 4.1 (Physical mean residue)**.** *Choose $r = e^{-c_0\varepsilon}$ as in Lemma 2.1. The singular part of the localized chart is obtained by moving the real v-segment downward through $Y = 1$; the resulting coefficient is $+2\pi$. At that residue the denominator satisfies*

$$|1 - Z(t, s_\varepsilon)| \ge c_1(\varepsilon + \textstyle\sum_j t_j^2)$$

on a sufficiently small fixed shape ball.

Proof. Implicit differentiation of $\cos\phi = S - \cos(-\alpha + u)$ gives

$$b = \phi_u = \frac{\sin\alpha}{\sin\beta} > 0, \qquad d = -\tfrac{1}{2}\phi_{uu} = \frac{S_*(1 - \cos\alpha\cos\beta)}{2\sin^3\beta} > 0.$$

At the radial point,

$$\phi = \beta - \mathrm{i}A_0\varepsilon + bu - du^2 + O(\varepsilon^2 + \varepsilon|u| + |u|^3), \quad A_0 = \frac{2\sin\theta_* - c_0\sin\alpha}{\sin\beta}.$$

Choose c_0 so that $A_0 > c_0$, in addition to global residue admissibility. On a rectangle $|\Re v| \le \delta$, $-h_0 \le \Im v \le 0$, take the fixed shape support $\sum t_j^2 < h^2$ with $dh^2 \ll b\delta$. Take h_0 sufficiently small but fixed and $Nc_0\varepsilon < h_0$. The Y-pole is $v = -\mathrm{i}Nc_0\varepsilon$. The Z-pole is above the real mean

line in the local model; throughout this rectangle a downward mean motion increases its damping. The side phases are separated by $b\delta + O(h^2 + h_0^2)$, and the bottom has fixed positive damping. All pair denominators remain nonzero since $z_0^2 \neq 1$. Thus the three displaced sides, with any fixed number of s-derivatives, are bounded. A shape cutoff independent of v is legitimate here; smoothing the hard mean endpoints is supported on the separated sides.

Orientation and residue. The downward closure is clockwise. Since $\partial_v(1 - Y) = -\mathrm{i}$ at the pole, the orientation and derivative give $(-2\pi\mathrm{i})/(-\mathrm{i}) = +2\pi$. There each $y_j = e^{\mathrm{i}(-\alpha+t_j)}$ lies exactly on the unit circle. On this compact regular arc, $-\log|Z| \geq cN\varepsilon$. The real phase $g_\varepsilon(\theta) = \Re \arccos(S(s_\varepsilon) - \cos\theta)$ has $g_\varepsilon'' \leq -c < 0$, and $Ng_\varepsilon(-\alpha) = N\beta + O(\varepsilon^2)$. To turn curvature into a denominator estimate, apply Taylor's integral remainder on $\sum t_j = 0$. It supplies a phase separation $c\sum t_j^2$ when that sum exceeds a sufficiently large multiple of ε^2. Below this scale the modulus separation controls it. Shrinking the ball prevents phase wrapping. Combining the two separations proves the asserted nonlinear bound. □

Lemma 4.2 (Nonzero shape period)**.** *Let*

$$a_0 = \frac{N^2 - 1}{2}, \quad Q = \frac{2N\sin\theta_*}{\sin\beta}, \quad \mathscr{C}_N = \frac{1}{\sqrt{N}} \int_{\{\omega\in\mathbb{R}^N:\, \sum\omega_j = 0,\ |\omega|=1\}} \Delta(\omega)^2 \, \mathrm{d}\sigma(\omega) > 0.$$

Then

$$J_\beta = \int_{\mathbb{R}^{N-1}} \frac{\Delta(\tau)^2 \, \mathrm{d}\tau_1 \cdots \mathrm{d}\tau_{N-1}}{(Q - \mathrm{i}d\sum_j \tau_j^2)^{k+1}} = \frac{\mathscr{C}_N}{2} Q^{-1/2} (-\mathrm{i}d)^{-a_0} B(a_0, 1/2) \neq 0. \tag{4.1}$$

Here $\tau_N = -\sum_{j<N} \tau_j$, and the power is obtained by continuation from a quadratic coefficient with positive real part.

Proof. The shape integral is evaluated on the zero-sum hyperplane. Its volume normalization and the continuation of the quadratic coefficient both matter for the complex amplitude.

Polar coordinates in the zero-sum hyperplane give radial numerator R^{N^2-2}; the Euclidean-to-coordinate volume factor is $1/\sqrt{N}$. The substitution $u = R^2$ gives Euler's beta integral. Continuation to the coefficient $-\mathrm{i}d$ is dominated by a constant times $R^{N^2-2}(1+R^2)^{-N^2/2}$, whose tail is $O(R^{-2})$. All factors on the right of (4.1) are nonzero. □

The leading local density and pole derivative are

$$K_\beta = \frac{2}{N!} \, 2^{-N(N-1)} (2\pi)^{-N} \sin(\beta)^{-N^2}, \qquad D_s = -\frac{\mathrm{i}NS'(s_*)}{\sin\beta} = \frac{2Ne^{-\mathrm{i}\theta_*}\sin\theta_*}{\sin\beta}.$$

Indeed every pair has leading factor $-(u_i - u_j)^2/(4\sin^2\beta)$, and each residue-angle measure has leading factor $y_0 z_0/(2\pi\mathrm{i}\sin\beta)$. Since N is even and $y_0^N z_0^N = 1$, their combined phase is $\mathrm{i}^{-N^2} = 1$, giving $K_\beta > 0$.

Shape rescaling and the two chart coefficients. After the mean residue, put $t = \sqrt{\varepsilon}\,\tau$. The k-th derivative has leading coefficient

$$L_\beta = 2\pi K_\beta (-1)^k k! D_s^k J_\beta \neq 0$$

times $\varepsilon^{-1/2}$. The bound in Lemma 4.1 dominates the rescaled integrand by $C\Delta(\tau)^2/(1+|\tau|^2)^{k+1}$. Lower pole orders vanish after multiplication by $\sqrt{\varepsilon}$. This majorant is for the constrained shape integral obtained after taking the mean residue. The sine powers in $K_\beta D_s^k Q^{-1/2} d^{-a_0}$ sum to

$-N^2 - k + 1/2 + 3a_0 = 0$. Hence $L_\alpha = L_\beta$ for the swapped ordered chart, including the phase. Exchanging all x's with all y's in (2.1) also preserves orientation for even N and checks this equality before asymptotics.

Lemma 4.3 (Complement of the two first charts)**.** *At $N = N_0$ all other localized charts of the original double contour have bounded derivatives of every fixed order on the exterior radial approach.*

Proof. First classify the possible nonnegative relations among singular-factor gradients. Away from the two surviving configurations, a separating direction then gives an integrable auxiliary-parameter bound after repeated integration by parts.

The local singular-factor gradients in [10] are the two all-ones product vectors, pair vectors, and the dispersion vectors with entries $(\Im x_j, \Im y_j)$. A nonnegative zero combination cannot contain a pair vector: cancellation of its two positive entries would require two negative imaginary parts, whereas reciprocal coordinates have opposite imaginary parts. If both product vectors occur, every x_j, y_j is lower, the ratios $\Im x_j / \Im y_j$ are equal, and the dispersion relation forces all x_j's equal and all y_j's equal. Explicitly, with $x = e^{-\mathrm{i}A}, y = e^{-\mathrm{i}B}$, $A, B \in (0, \pi)$, differentiation along $\cos A + \cos B = S_* > 0$ gives

$$\frac{\mathrm{d}}{\mathrm{d}A} \frac{\sin A}{\sin B} = \frac{S_*(1 - \cos A \cos B)}{\sin^3 B} > 0.$$

Thus equality of these ratios determines the lower dispersion pair. Both common values are N-th roots of unity. Theorem 3.1 leaves exactly the two charts. If only one product vector occurs, the other coordinates are all $+1$ or all -1. The former would make $e^{-\mathrm{i}\theta_B}$ an N-th root of unity; the latter requires a real cosine $S_* + 1 > 1$. Both are excluded.

Uniform bound on the complement. Fix a smooth weight supported away from these two configurations and cover its support by finitely many compact localized pieces. On each piece choose a real separating direction e and $\delta_* > 0$ with $e \cdot v_i \geq \delta_*$ for every active singular-factor vector v_i. Shrink the piece once so that this separation persists uniformly for the nearby damped contours.

Represent its ℓ potentially singular factors by damped positive-half-line exponential integrals, and put $R = \sum_{i=1}^{\ell} \xi_i$ for their nonnegative auxiliary parameters. The exponent g satisfies $\Re g \leq 0$, $|e \cdot \nabla g| \geq cR$, and every fixed angular derivative of g is $O(R)$. An s-derivative of order j, taken at fixed contour radius, adds a polynomial in the auxiliary parameters of degree at most j, with uniformly bounded smooth coefficients. On $R \geq 1$, integrate by parts $q = \ell + j + 1$ times using

$$\mathcal{L} = \frac{e \cdot \nabla}{e \cdot \nabla g}, \qquad \mathcal{L} e^g = e^g.$$

The transpose of this operator gains one power R^{-1} per iteration, including derivatives of its coefficients and the localized weight. The angular integral is therefore bounded by CR^{j-q}. The auxiliary radial volume is $R^{\ell-1}\,\mathrm{d}R$, so the remaining tail is bounded by $C \int_1^\infty R^{-2}\,\mathrm{d}R$. The region $R \leq 1$, and pieces with no singular factor, are bounded directly. The finite cover supplies common constants for each fixed N, j. In particular, the bound applies to the complementary weight at the selected Nickel point, where the active vectors are separated from zero on its support. □

Lower-order radial bounds from the original contour. Fix a positive even $n < N_0$, a finite derivative order j, and s_* as above. There are $\varepsilon_{n,j}, C_{n,j} > 0$ such that

$$\left|\partial_s^j T_n(s; r_\varepsilon)\Big|_{s=s_\varepsilon}\right| \le C_{n,j}, \qquad 0 < \varepsilon < \varepsilon_{n,j}, \qquad r_\varepsilon = e^{-c_0\varepsilon}.$$

Here $T_n(s; r)$ is the literal offsite double integral (2.1); each derivative holds $r = r_\varepsilon$ fixed. Choose the fixed $c_0 > 0$ small enough that $r_\varepsilon^{-1} - r_\varepsilon < \Im S(s_\varepsilon)$, with strict slack on a $c\varepsilon$ complex neighborhood of each evaluation point.

The same active-gradient classification as in Lemma 4.3, now with n coordinates, proves this bound. Pair coefficients again vanish. If both product coefficients are positive, the common lower coordinates x, y obey $x^n = y^n = 1$ and $\Re x + \Re y = S_*$, a Nickel equation of order n. If only the x-product coefficient is positive, every $y_j = 1$: the alternative -1 would give $\Re x_j = S_* + 1 > 1$. All lower x_j's then have real part $S_* - 1$, hence coincide; their product is one. Since 1 is itself an n-th root, this too is a Nickel equation. The single y-product case is its exchange. If neither product occurs, each dispersion coefficient vanishes: its two imaginary coordinates cannot both vanish when $0 < S_* < 2$. Theorem 3.1 excludes all the resulting Nickel equations for $n < N_0$.

With every possible Nickel configuration excluded, the active vectors at each point of the entire $2n$-torus have convex hull separated from zero. Choose a finite smooth partition and positive separating directions at the limiting torus, before varying ε, r, s. Inactive denominators and their fixed jets are bounded on these fixed charts. Continuity preserves the separating margins for nearby (r, s). On the specified radial family all exponential factors have nonpositive real exponent, including dispersion damping. For ℓ active factors a fixed s-derivative costs at most R^j, while $q = \ell + j + 1$ integrations by parts give the integrable tail $R^{\ell-1+j-q}\,\mathrm{d}R = R^{-2}\,\mathrm{d}R$, exactly as in the lemma. The small-R part is bounded directly. This proves the bound for the original offsite contour, including its inverse-product numerator, normalized measures and factorial. It does not transfer an external full-site statement to a different observable.

The admissible contours give the same holomorphic T_n locally in s, so these fixed-radius derivatives are its derivatives. The onsite coefficient is bounded by the same argument with both global product factors omitted; Appendix F's identity $C_n^{\mathrm{std}} = f_{00}^{(n)} + 2T_n$ therefore also gives the corresponding full-site radial bound. The constants may depend on n, j, s_*. For $n < N_0$ and $j \le k$, take the minimum of the finitely many positive radial thresholds.

Theorem 4.4. *At the selected point,*

$$T_{N_0}^{(k)}(s_\varepsilon) = 2L_\beta\,\varepsilon^{-1/2} + o(\varepsilon^{-1/2}), \qquad L_\beta \neq 0.$$

Every derivative of order $0 \le j < k$ of this first term is bounded on the radial approach. Every even $N < N_0$ has bounded derivatives there as well.

Proof. We now assemble the localized contributions. The only unbounded leading terms are the two mean residues; their equality of phase makes them add, while the complement and the lower particle orders remain bounded.

The compatible y-only partition gives the two reduced local integrals and a complement bounded by Lemma 4.3. Replacing their smooth mean cutoffs by hard segments adds only bounded edge terms. The two actual mean residues therefore add with the displayed coefficients. For lower derivatives the constrained radial density at $\varepsilon = 0$ has exponent $N^2 - 2j - 4 \ge 0$, so it is integrable. The displaced sides are bounded as before. Lower even orders are bounded by the preceding radial estimate, which uses Theorem 3.1. □

5 Exact contour decomposition and quantitative requirements

The tail proof begins with an identity, before making any estimates. A weighted contour deformation separates a selected final contour, a weighted original contour, and the current generated by the weight. After differentiation these pieces are divided into particle-number and current-parameter windows. The four bounds at the end of this section are the targets used in the final summation. For the remainder fix (p, a, b), $N_0 = 2p$, and $k = N_0^2/2 - 1$. Set $H = \log(1/\varepsilon)$ and choose

$$0 < \alpha_0 < \frac{1}{2(k+2)}, \qquad \lambda_* = \varepsilon^{\alpha_0}. \tag{5.1}$$

Neither N_0 nor k varies with ε. Let $\Omega_{N,s}$ be the reduced holomorphic form in (2.3), including its $1/N!$. Choose smooth periodic functions p, m, a with values in $[0, 1]$ as follows, in angular representatives $-\pi < \theta \le \pi$. The closed support of m is contained in a small neighborhood of $-\theta_B$ within $(-\pi, 0)$; $m = 1$ on an open neighborhood of the closed branch-cutoff supports in Lemma 7.1. Thus it is one on the supports of all their finite jets as well. Choose a small fixed endpoint width $\delta > 0$ and take

$$a = 0 \text{ off } [\delta, \pi - \delta], \qquad a = 1 \text{ on } [2\delta, \pi - 2\delta].$$

Make δ small enough that the reflection of $\operatorname{supp} m$ lies in the interior of the latter interval. Take p with closed support in $(0, \pi)$, equal to one on an open neighborhood of $[\delta, \pi - \delta]$. Smooth step functions give all these choices simultaneously. Their widths and the deformation strength are fixed in the order of Appendix E, before $\varepsilon \to 0$. The supports of p, m are disjoint, the upper inverse branch is inside $a = p = 1$, and $\operatorname{supp} a$ and $\operatorname{supp} a'$ have positive distance from both endpoints. In particular a named upper angle has $\sin\theta \ge \sin\delta > 0$.

Define, with fixed deformation strength $\tau > 0$,

$$P = \sum_i p(\theta_i), \quad h_i = -2\tau p(\theta_i) + \frac{\tau P}{2N} m(\theta_i), \quad y_i(\lambda, \theta) = r e^{\mathrm{i}\theta_i + \lambda h_i}, \qquad \psi(\theta) = \prod_i (1 - a(\theta_i)).$$

Write $U_{\lambda,s} = \Phi_\lambda^* \Omega_{N,s}$.

Proposition 5.1 (Exact weighted retraction)**.** *For exterior s in its admissible neighborhood,*

$$T_N = F_N + K_N + S_N, \tag{5.2}$$

$$F_N = \int (1 - \psi) U_{1,s}, \qquad K_N = \int \psi U_{0,s},$$

$$S_N = -2\mathrm{i}\tau \int_0^1 \sum_q \int (\partial_{\theta_q} \psi) U_{\lambda,s} \, \mathrm{d}\lambda. \tag{5.3}$$

The current has one named index q in $\operatorname{supp} a'$, with $p_q = 1$, $p_q' = m_q = m_q' = 0$; all other indices belong to the enlarged lower support $K = \operatorname{supp}(1 - a)$.

Proof. We have $\sum_i h_i \le -3\tau P/2$, so $|Y| < 1$. An inward radial motion on an upper angle and an outward radial motion on a lower angle both increase $\Im W$: $\partial_{\log|y|} \Im W = -\cosh(\log|y|) \sin\theta$. Thus all z_i's stay on the interior sheet. Neither $1 - Z$ nor a denominator $1 - z_i z_j$ vanishes. The canceled pair expression has no remaining $1 - y_i y_j$ pole. This proves legality for the full closed homotopy.

Jacobian and current coefficient. Normalize the angular Jacobian row i by y_i. Its matrix is

$$M_{ij} = d_i\delta_{ij} + \frac{\lambda\tau}{2N} m_i p'_j, \qquad d_i = \mathrm{i} + \lambda\left(-2\tau p'_i + \frac{\tau P}{2N} m'_i\right).$$

Because $m_i p'_i = 0$, $\det M = \prod_i d_i$; all factors are nonzero and its absolute value is bounded by C^N. At a named q, both the row and column of M are zero off the diagonal and $d_q = \mathrm{i}$. Replacing angular column q by the homotopy velocity multiplies the determinant by $(-2\tau)/\mathrm{i} = 2\mathrm{i}\tau$. Cartan's formula on the closed torus gives

$$\partial_\lambda \int \psi U_{\lambda,s} = -2\mathrm{i}\tau \sum_q \int (\partial_{\theta_q}\psi) U_{\lambda,s}.$$

The unweighted integral is homotopy invariant. Integrating this identity and separating the selected part proves (5.2). □

The weighted deformation includes the current term: an individual weighted chart changes under the deformation. To differentiate the identity, take all s-derivatives over the fixed interval $[0,1]$, and then split the resulting density at λ_*. This order of operations introduces no derivative of the split endpoint.

Proposition 5.2 (Ultra-high orders)**.** *For fixed s_* and $0 \le j \le k$, there are constants $C_0, C_1, c > 0$, independent of N, ε, such that*

$$|T_N^{(j)}(s_\varepsilon)| \le C_j \varepsilon^{-j-2} \left(\frac{C_1(H+1)}{\sqrt{N}}\right)^N.$$

For a sufficiently large fixed C_H, the sum over even $N \ge C_H(H+1)^2$ is $o(\varepsilon^A)$ for every fixed real A.

Proof. At very large particle number, factorial suppression alone is strong enough. The argument bounds one matched pair in the Pfaffian, counts matchings, and then applies Cauchy on the original parameter disk.

On the original $c\varepsilon$ disk, each root lies in the lower unit half-disk. The identity

$$|1 - zw|^2 - |z - w|^2 = (1 - |z|^2)(1 - |w|^2) + 4\Im z \Im w$$

and the Schur identity give $|\operatorname{Pf} A(z)| = \prod_{i<j} |(z_i - z_j)/(1 - z_i z_j)| \le 1$. The product numerator is bounded by C^N and the two global denominators below by $c\varepsilon$. At the two simple inverse-dispersion branch angles, $|R(re^{\mathrm{i}\theta}; s)|^2 \le C/(\varepsilon + \operatorname{dist}(\theta, \{\pm\theta_B\}))$. Its squared L^2 norm is $O(H+1)$. The y-pair kernel has a majorant depending only on $\theta_i + \theta_j$, with L^1 norm $O(H+1)$. Circular convolution and Cauchy–Schwarz therefore bound one matched weighted pair integral by $C(H+1)^2$.

The y-Pfaffian has $N!/[2^{N/2}(N/2)!]$ matchings. Each matching factorizes after these majorants. Dividing by $N!$ and using $(N/2)! \ge (N/(2e))^{N/2}$ proves the undifferentiated bound. Cauchy on the same disk adds $j!(c\varepsilon)^{-j}$. Choose C_H so that the bracket is at most $1/2$ at the threshold. The summed bound is then $C_j \varepsilon^{-j-2} 2^{-C_H(H+1)^2}$, which has the stated decay. □

For $N < C_H(H+1)^2$, denote the already differentiated, angularly integrated current by

$$\mathcal{J}_{N,j}(\lambda) = -2\mathrm{i}\tau \sum_q \int (\partial_{\theta_q}\psi)\, \partial_s^j U_{\lambda,s}\big|_{s=s_\varepsilon},$$

$$B_{N,j} = |K_N^{(j)}(s_\varepsilon)| + \left|\int_0^{\lambda_*} \mathcal{J}_{N,j}(\lambda)\, \mathrm{d}\lambda\right|.$$

The remaining work is organized by the following four quantitative requirements. The first two sum the selected contour and the large current parameter; the last two control the original contour and the small current parameter in their respective particle windows:

$$\sum_N |F_N^{(j)}| \le \exp(C_j \log^2(H+2)), \tag{5.4}$$

$$\sum_N \left| \int_{\lambda_*}^1 \mathcal{J}_{N,j}(\lambda)\, \mathrm{d}\lambda \right| \le C_j \varepsilon^{-\alpha_0(j+2)}, \tag{5.5}$$

$$B_{N,j} \le C_j^N \varepsilon^{-j-2} e^{-\kappa N^2} \qquad (N \ge D\sqrt{H}), \tag{5.6}$$

$$B_{N,j} \le \exp(C_j N \log(N+1))(H+N)^{C_j} \qquad (N_0+2 \le N < D\sqrt{H}). \tag{5.7}$$

The sums are over even tail orders $N > N_0$ below $C_H(H+1)^2$. These are statements with fixed point and j, followed by constants, followed by a single sufficiently small ε_0, and only then all orders in the indicated window.

6 The final selected contour

The selected contour contains an upper anchor moved a fixed distance inward. This anchor opens both global product gaps and protects the branch roots. Its parameter disk is therefore larger than the original radial disk. The proof uses that disk only after bounding the complete selected integral.

Proposition 6.1. *The bound* (5.4) *holds for the final selected integral.*

Proof. We first obtain the protected disk, then estimate the two Pfaffians, and finally differentiate the complete integral. The pointwise group assignments used in the absolute estimates are never differentiated.

The anchor and its protected disk. Telescope $1-\psi$. A named selected index has $p_q = 1$, hence $P \ge 1$. At $\lambda = 1$, this index is shifted inward by 2τ. The endpoint separation above gives $\Im W_q \ge \sinh(2\tau)\sin\delta$ at the radial center. The interior root therefore stays in a strict subdisk, uniformly on the selected support. It supplies a fixed Z-gap, and $\sum h_i \le -3\tau P/2$ supplies a fixed Y-gap. Every true lower branch index is on $m = 1, p = 0$ and has $\Im W \ge cP/N$, so its residue factor is at most $C\sqrt{N}$. Other roots are in a compact regular set or on the shifted upper branch. The one-body product and pullback Jacobian cost $C^N N^{N/2}$.

These statements continue to the common disk $|s - s_\varepsilon| \le c/N$: the lower branch margin is c'/N, compact roots move $O(1/N)$, and the fixed anchor absorbs the total compact-root inflation $\exp(CN(c/N))$ after the disk constant is made small. The integral on this disk is holomorphic because the contour weights are fixed in s. Only the original undeformed representation needs the smaller $c\varepsilon$ disk.

Root-pair contraction. Make the root assignments pointwise in s to bound the absolute integral. Cauchy's inequality will apply to the complete integral. Let M be the number of indices outside the true lower branch core. Split their compact z-images into a strict interior group and a compact lower-arc group. On this disk, same-group compact Schur pairs have modulus at most $q < 1$; cross compact pairs are at most $1 + C/N$; branch/compact pairs are also at most $1 + C/N$; branch/branch pairs are at most one.

The branch–compact pairs require a uniform estimate on the whole protected disk. The following comparison holds the branch root fixed and moves only the compact root. On the

entire protected disk a true branch root w remains in the closed lower unit disk: its W-image has strictly positive imaginary part by the P/N margin. The compact radial-center root z_0 belongs to a fixed compact subset of that same half-disk excluding ± 1. Thus for some $g > 0$, either $|z_0| \le 1 - g$ or $\Im z_0 \le -g$. In both cases $|1 - z_0 w| \ge g$. For the second case this follows directly from

$$|z_0|^2 |1 - z_0 w|^2 = (|z_0|^2 \Re w - \Re z_0)^2 + (|z_0|^2 \Im w + \Im z_0)^2.$$

The compact root moves to z by at most L/N; reduce the disk constant so this is at most $g/2$. With $A(z, w) = (z - w)/(1 - zw)$,

$$A(z, w) - A(z_0, w) = \frac{(z - z_0)(1 - w^2)}{(1 - zw)(1 - z_0 w)}, \qquad |A(z, w)| \le 1 + \frac{4L}{g^2 N}.$$

Only the compact root was displaced in this comparison: w is held at its actual disk-point value. Hence no square-root branch motion enters the estimate. Compact roots on both transition supports are covered by the two compact groups, and no perturbed compact root is assumed to lie in the unit disk. For compatibility with the existing budget we may weaken this bound to $1 + C/\sqrt{N}$ in the next display. Since the same-group count is at least $M^2/4 - M/2$,

$$|\operatorname{Pf} A(z)| \le \exp(CN + CM\sqrt{N} - aM^2) \le C_1^N e^{-a_1 M^2}.$$

The last inequality uses $CM\sqrt{N} \le (a/2)M^2 + C^2 N/(2a)$; the mixed inflation cannot be dropped without this step.

Angular-pair matching. Every y-pair with a true branch index is bounded. Indeed at reciprocal angles its reflected upper partner has $p = 1$, and the negative shift -2τ dominates the possible positive lower shift $\tau P/(2N) \le \tau/2$. For two compact angles write $d = \operatorname{dist}(\theta_i + \theta_j, 2\pi\mathbb{Z})$. At exact reciprocity $h_i + h_j \le 0$ for every $P/N \in [0, 1]$; smoothness gives $h_i + h_j \le Ld$. For $d \le c\varepsilon$ the negative base $2 \log r$ bounds the modulus gap below by $c'\varepsilon$; for larger d the phase bounds it below by $c'd$. Thus

$$|A(y_i, y_j)| \le \frac{C}{\varepsilon + d}.$$

This majorant is independent of all other angles, including the coupled scalar P. Its matched two-angle integral is $O(H + 1)$.

Write the y-Pfaffian matrix as $A_0 + E$, where E has entries only between compact indices and A_0 has bounded entries. Expand in an even compact subset of size $2r$. The remaining Pfaffian minor is at most $C^N N^{(N-2r)/4}$ by Hadamard. There are at most N^{3r} coarse choices of subset and matching; their disjoint kernels have integral at most $(C(H + 1))^r$. As $r \le M/2$, a sufficient bound is

$$\int |\operatorname{Pf} A(y)| \le C^N N^{N/4} (CN^3 (H + 1))^M.$$

Factorial suppression and differentiation. Combine the two Pfaffian bounds with $N! \ge (N/e)^N$. For each assignment this gives

$$|F_{N,\text{assignment}}(s)| \le C^N N^{-N/4} (CN^3 (H + 1))^M e^{-a_1 M^2}.$$

At most C^N assignments and a polynomial number of named indices are absorbed by increasing C. For $N < C_H (H + 1)^2$, completing the square in M costs $\exp(C \log^2(H + 2))$. This gives the same bound for the complete $|F_N(s)|$ throughout the common disk. Its fixed angular weights

and the uniform integrable majorant above make F_N holomorphic on a neighborhood of that closed disk, after decreasing c once. Apply Cauchy to this complete integral:

$$|F_N^{(j)}(s_\varepsilon)| \le j! \left(\frac{N}{c}\right)^j \sup_{|s-s_\varepsilon|\le c/N} |F_N(s)|.$$

No analytic dependence of the pointwise assignments is needed. Since $\sum_N C^N N^{-N/4+j}$ converges, this proves (5.4). □

7 The original contour and the all-branch region

The original lower support needs a different estimate from the selected contour. We first build a partition whose support survives every cutoff derivative. Complete pair factors then provide Gaussian decay in the particle number. For the remaining intermediate orders, the all-branch region is divided into a small nonresonant core and its complement.

Lemma 7.1 (Fixed support partition)**.** *The original K-integral has a finite smooth partition into an all-lower-branch region and regions of the following types: a compact left root with a modulus gap, a separated compact/true-branch pair, and an all-compact right region. The partition uses at most C^N labels, and every finite cutoff jet retains its named anchor and its stated support. Each named small-λ Stokes current has the same three compact/branch types, with no all-branch sector.*

Proof. The partition must retain its support properties under differentiation. We therefore construct nested branch cutoffs first, place the periodic seam where the weighted integrand vanishes, and then distinguish the current by its named upper index.

Use $u = \theta + \theta_B$ near the lower branch. Fix a small $\delta_B > 0$ and a large fixed L. Take $\chi_B = 1$ for $|u| \le \delta_B/2$, supported in $|u| \le \delta_B$. Take $\chi_{\mathrm{in}} = 1$ for $|u| \le r_0 = \delta_B/(2L^2)$, supported in $|u| \le 2r_0$. The deformation function m is one on a neighborhood of these supports. Split $1 = \prod \chi_B + (1 - \prod \chi_B)$ and telescope the second term. It has a named q with $|u_q| \ge \delta_B/2$, including on the support of every derivative of that named factor. Expand the other coordinates with $1 = \chi_{\mathrm{in}} + (1 - \chi_{\mathrm{in}})$.

Periodic labels with an inactive seam. To make the left/right labels periodic, choose an open arc J whose closure lies inside the upper $a = 1$ plateau. Cut the circle at a point of J and unwrap the angle on its complement. Near $-\theta_B$, use $u = \theta + \theta_B$ and a smooth right step which is zero for $u \le -r_0/4$ and one for $u \ge r_0/4$. Away from this local transition keep its value constant on each side, and return smoothly from one to zero inside J. This defines a smooth periodic right label; the left label is its complement.

At the branch transition every derivative of the products with $1 - \chi_{\mathrm{in}}$ vanishes, since $\chi_{\mathrm{in}} = 1$ there. The additional transition inside J is also harmless: $1 - a$ vanishes there with all its jets. Thus the full K-weight and every current weight $\partial_{\theta_q}\psi$, together with all their jets, vanish whenever any coordinate lies in J. For the named current coordinate this follows from $a' = 0$; for every other coordinate it follows from its factor $1 - a$. On the support of any relevant weighted jet the labels therefore select the required side. On the transition of χ_{in} they are constant. Every effective compact label excludes a fixed branch neighborhood. If a true-branch label occurs, the outer q and a true-branch j satisfy $|b_j| \ge 4|b_q|$ after choosing L large, unless a left root already supplies a modulus gap. With no branch label, all roots are compact; either one is left or all are

right. Finite product differentiation creates at most $N^{O(j)}$ terms and preserves the supports just described.

Current supports. For the current fix its named index q_S in $\operatorname{supp} a'$. Split its two upper endpoint arcs; this split can be smooth where the current weight vanishes. An upper-left q_S gives a fixed root modulus gap. For an upper-right q_S, expand every other index with $\chi_{\text{in}} + (1 - \chi_{\text{in}})$ and the compact left/right labels. A compact left index again gives a modulus gap. Otherwise, if the true-branch label set J_B is nonempty, use q_S and the fixed index $\min J_B$ as the mixed pair; if it is empty all indices are compact right. These alternatives are exhaustive, including when all $N - 1$ other indices are true branch. The named upper index prevents an all-branch sector.

The slope of q_S is uniformly bounded on a preliminary endpoint neighborhood separated from the upper branch, independently of subsequently shrinking the slivers. Choose the nested branch margin so that its limiting slope dominates this bound by more than four. Lemma 7.4's deformed estimates preserve $|b_{q_S}/b_j| \le 1/4$ after the final choice of ε_0. The named index remains on $p = 1, m = 0$, and every true branch on $p = 0, m = 1$, on neighborhoods of all cutoff-jet supports. The vanishing on J proved above applies also to the current. Thus no cutoff derivative creates a new support type. □

Lemma 7.2 (Pair suppression at radial centers)**.** *After the fixed branch margins are chosen, the endpoint slivers and deformation strength can be chosen so that at $s = s_\varepsilon$, on K^N and on each Stokes support with one named q,*

$$\left| \prod_{i<j} P_{ij} \right| \le C^N e^{-\kappa N^2}, \qquad \kappa > 0.$$

This is uniform for $0 \le \lambda \le 1$, all permitted occupancies $P/N \in [0,1]$, and sufficiently small ε. More precisely, lower indices can be assigned to three fixed covering groups with same-group bound $q' < 1$ and cross-group bound $1 + \eta$, where $\log(1+\eta) < -\log(q')/4$. Pairs containing the named Stokes index are bounded by $C_S \ge 1$. For every set E of at most j pairs, these bounds imply

$$\prod_{\substack{i<l \\ (i,l)\notin E}} |P_{il}| \le C_j^N e^{-\kappa N^2}.$$

Fixed pair jets are bounded in regular branch coordinates and compact angular coordinates.

Proof. **Fixed groups and limiting equality loci.** Write $b = \theta_B$, $y_B = e^{-\mathrm{i}b}$, and $t_0 = S_* + 1 - \sqrt{(S_*+1)^2 - 1} \in (0,1)$. Choose a fixed $w < \min(b, \pi - b)/3$, small enough for the regular lower-branch chart, with $[-b - w, -b + w]$ inside the $m = 1$ neighborhood. Choose $0 < \ell_L, \ell_R < w$. The closed limiting groups are

$$B = [-b - w, -b + w], \quad L = [-\pi, -b - \ell_L], \quad R = [-b + \ell_R, 0].$$

They cover the lower semicircle with overlaps. Assign each overlap by any fixed rule. Add the upper-left sliver $[\pi - 2\delta, \pi]$ to L and the upper-right sliver $[0, 2\delta]$ to R, identifying π and $-\pi$. The named current index lies in one of the two transition arcs. All cutoff jets retain these closed supports. The branch group contains neither endpoint; otherwise $|P(y_B, 1)| = |P(y_B, -1)| = 1$ would prevent strictness.

At the limit $W_0(\theta) = S_* - \cos\theta$ and $W_0 + 1 \ge S_* > 0$. On $[-\pi, -b]$ the roots are real in $[t_0, 1]$; on $[-b, 0]$ they are $e^{-\mathrm{i}v}$, $0 \le v \le b$. The identity

$$|1 - zw|^2 - |z - w|^2 = (1 - |z|^2)(1 - |w|^2) + 4\Im z \Im w$$

and its unit-circle sine version give $|P| \le 1$. Apart from the removable configurations described below, equality requires an endpoint in the y-pair and equality in the z-factor. Its complete loci, with either ordering, are $y_i = 1,\ \theta_j \in [-\pi, -b]$, and $y_i = -1,\ \theta_j \in [-b, 0]$. Common-endpoint pairs have canceled value zero. No equality locus lies in a same-group square. On L^2 the roots stay in a strict real subinterval of $(0,1)$; on R^2 they stay on a compact lower arc away from the real axis. On B^2 the y-factor is bounded by

$$\frac{\sin w}{\min\{\sin(b-w), \sin(b+w)\}} \le \frac{1}{2\cos w} < 1.$$

The canceled extension below covers its double branch. Thus the three compact same-group suprema have a maximum $q < 1$. Cross-group suprema may equal one.

Local continuation and the complete pair. Use thin simply connected neighborhoods of the compact real $W_0 > 1$ intervals and of the compact intervals inside $(-1, 1)$, avoiding both branch points. On the former choose the root real in $(0,1)$; on the latter choose the continuation of $e^{-\mathrm{i}\arccos W_0}$. These local roots may leave the unit disk. Near the lower branch use $\phi = \arccos W$ on the fourth-quadrant sheet for $\Im W \ge 0$. The implicit-function theorem gives the even analytic inverse $y_s(\phi) = G_s(\phi^2)$ near $(s_*, 0)$, since $\partial_y W(y_B; s_*) \ne 0$. In this neighborhood

$$y_s(\phi_i) - y_s(\phi_l) = (\phi_i - \phi_l)(\phi_i + \phi_l) H_s(\phi_i, \phi_l), \qquad 1 - e^{-\mathrm{i}(\phi_i + \phi_l)} = (\phi_i + \phi_l) J(\phi_i + \phi_l),$$

where $J(0) = \mathrm{i}$. Substitution into (2.4) gives $P_{il} = (\phi_i - \phi_l)^2 A_s$, with analytic bounded fixed jets and $A_{s_*}(0,0) = -1/(4\sin^2 b)$. This resolves the whole local reciprocal-root hypersurface, without dividing by a collision zero.

At the limiting lower support $1 - z_i z_l = 0$ forces both roots to be 1 or both -1. The latter is excluded by $W_0 + 1 \ge S_*$; the former is the resolved lower double branch. On the closed compact complement of a smaller double-branch neighborhood, the denominator therefore has a positive minimum μ. Shrink its neighborhoods to retain $\mu/2$. The regular and rational expressions agree on overlaps by (2.4). This finite cover supplies continuous complete pairs and bounded fixed jets in regular branch and compact coordinates. No globally glued square-root domain is needed. The apparent $1 - y_i y_l$ pole was already canceled; the upper inverse branch, which would make a genuine pole with the lower branch, is excluded by the selector plateau.

For a named right limiting endpoint root $\xi = e^{-\mathrm{i}b}$, $|1 - \xi z| = |e^{\mathrm{i}b} - z| \ge \sin b$ for every limiting lower root. For the left endpoint it is at least $1 - t_0$. Thus a positive named margin $\mu_S = \min(\sin b, 1 - t_0)$ persists, with $\mu_S/2$ on sufficiently small neighborhoods. Boundedness of the canceled numerator supplies $C_S \ge 1$; no named contraction is needed.

Uniform support transfer and reserved slack. Project each sliver to its endpoint and keep each lower angle fixed; denote the projected angle by $\bar\theta_i$. For $\rho = P/N \in [0,1]$ and $0 \le \lambda \le 1$,

$$v_i = \log|y_i| = -c_0\varepsilon + \lambda\tau(-2p_i + \rho m_i/2), \qquad |v_i| \le c_0\varepsilon + 5\tau/2.$$

Let L_S bound $|S'|$ on a fixed parameter neighborhood. Uniformly in this entire parameter box, hence for every actual coupled occupancy and every N,

$$|y_i - e^{\mathrm{i}\bar\theta_i}| + |W_i - W_0(\bar\theta_i)| \le C(\varepsilon + \tau + \delta) + L_S|s - s_\varepsilon|.$$

Compact-root changes are Lipschitz. Near the branch, the factorization $\phi = \sqrt{2(1-W)}h(1-W)$, with analytic h, gives a one-half Holder bound on the chosen closed half-plane. The regular

pair is Lipschitz in (s, ϕ_i, ϕ_l); on the compact complement the denominator margin gives the same property in (y_i, y_l, z_i, z_l). A finite cover therefore gives one uniform complete-pair modulus, bounded by $C\sqrt{\varepsilon + \tau + \delta + |s - s_\varepsilon|}$ in these neighborhoods. It applies whenever the branch-near roots stay on the designated sheet; the two disk proofs below verify this condition separately.

Choose $q < q_1 < q' < 1$, then $\eta > 0$ with $\log(1+\eta) < -\log(q')/4$. Choose the slivers and τ, and a preliminary ε_0, so the radial-center modulus is smaller than both $q_1 - q$ and $\eta/2$, while retaining the denominator neighborhoods. This gives same-group bound q_1 and cross-group bound $1 + \eta/2$; reserve $q' - q_1$ and $\eta/2$ for the disks. The real radial motions increase $\Im W$, so every radial-center root is interior. This transfer includes the slivers in the same-group estimate as well as the cross estimate.

The limiting branch/upper-sliver moduli for small $a > 0$ are

$$\frac{\sin((b+a)/2)}{\sin((b-a)/2)} = 1 + a\cot(b/2) + O(a^2), \qquad \frac{\cos((b-a)/2)}{\cos((b+a)/2)} = 1 + a\tan(b/2) + O(a^2).$$

An RR or LL pair can therefore exceed one if a compact-group boundary approaches the branch with a fixed sliver width. Choose the compact-group boundaries and contraction slack before the sliver width.

Let n be the number of lower variables: $n = N$ for K and $n = N - 1$ for a Stokes term. Assign each to one covering group. After at most j pairs have been removed, the remaining same-group count is at least $n^2/6 - n/2 - j$, while the remaining cross-group count is at most $n^2/3$. Apply the individual upper bounds directly to the remaining product:

$$(q')^{n^2/6 - n/2 - j}(1+\eta)^{n^2/3} \le (q')^{n^2/12 - n/2 - j}.$$

This remains valid if the lower bound on the count is negative. It involves no division by a removed factor, which may vanish. The named upper compact q has at most $N - 1$ additional pairs, each uniformly bounded: its z_q is separated from the inverse of a lower root, including the branch root. These pairs cost C_S^{N-1}. For $n = N - 1$ the displayed exponent is $N^2/12 - 2N/3 + 7/12 - j$; the linear correction is included in C_j^N. Thus $\kappa = -\log(q')/12$ works for both objects. The canceled double-branch expression is analytic in the regular coordinates, and the other pair denominators are separated on the fixed compact sets. This also proves the stated fixed-jet bounds. □

Lemma 7.3 (The original parameter disk)**.** *For a sufficiently small fixed $c > 0$, the original contour density and all K and Stokes transition densities are holomorphic on $|s - s_\varepsilon| \le c\varepsilon$. On the K and named Stokes supports, the following estimates hold:*

$$|1 - Y|,\ |1 - Z| \ge c'\varepsilon, \qquad \left|\prod_{i<j} P_{ij}\right| \le C^N e^{-\kappa' N^2}.$$

The integrated one-body majorants and contour Jacobian cost at most C^N, uniformly in N and $0 \le \lambda \le 1$.

Proof. The original parameter disk is used for the high particle orders. We establish the product gaps and an integrable one-body majorant, then transfer the complete-pair contraction to the disk on the stated supports.

The undeformed dispersion has $\Im W \ge c_2\varepsilon$. Every permitted radial motion increases this quantity, as in Proposition 5.1; reducing c preserves a fixed fraction of the margin on the disk. Each root is then in the lower unit half-disk with attenuation at least $c_3\varepsilon$. Thus the product gaps are bounded below by a constant times $\min(1, N\varepsilon)$, which is at least a constant times ε. The contour Y is independent of s.

Near the lower branch put $t = \lambda P/N$, $E = \varepsilon + t$, and $\delta = |s - s_\varepsilon| \le c\varepsilon$. On its true plateau, write $y = e^{v+\mathrm{i}\vartheta}$, where $v = -c_0\varepsilon + \tau t/2$ and $\vartheta = -\theta_B + u$. The exact identities

$$\Re W = \Re S - \cosh v \cos\vartheta, \qquad \Im W = \Im S - \sinh v \sin\vartheta$$

give, with $D_B = 1 - W$ and $a_B = \sin\theta_B > 0$,

$$|\Re D_B - a_B u| \le C(u^2 + \varepsilon^2 + t^2 + \delta), \qquad \Im W \ge c_b E - C_S\delta \ge (c_b/2)E.$$

Here the last inequality holds after reducing the disk constant c. All constants depend only on the fixed point and margins. In particular, the real branch-center displacement on this disk is $O(\varepsilon^2 + t^2 + \delta)$: the parameter perturbation cannot be omitted from the radial-center estimate.

An occupancy-independent majorant. Choose a fixed branch radius h and a fixed $K \ge 1$. If $|u| \le KE$, the imaginary margin gives $|D_B| \ge c_b(|u| + \varepsilon)/(2(K + 1))$. If $KE < |u| \le h$, then $u^2 \le h|u|$, $\varepsilon^2 + t^2 \le E^2 \le h|u|/K^2$, and $\delta \le cE \le c|u|/K$. Choose h, K, c so that $C(h + h/K^2 + c/K) \le a_B/2$, in addition to the previous margins. Then $|\Re D_B| \ge a_B|u|/2$ in this second region. These choices are independent of $N, \varepsilon, \lambda, P$. Since $W + 1$ stays separated from zero on the fixed branch arc and $R = z/\sqrt{W^2 - 1}$, both regions yield

$$|R(y; s)| \le \frac{C}{\sqrt{|u| + \varepsilon}}$$

uniformly on the full disk. The upper branch is absent from these supports; the remaining compact factors are bounded. This majorant is independent of P and has integral at most $4C\sqrt{h}$ on $|u| \le h$, so the product of majorants integrates to at most C^N. The inverse-product numerator and the Jacobian in Proposition 5.1 also cost C^N.

Transferring the pair bounds. For an explicit disk choice, let $b_0\varepsilon$ be the undeformed imaginary margin and let L_S be as in Lemma 7.2. Choose $L_S c \le b_0/4$. The monotone radial motions then leave $\Im W \ge 3b_0\varepsilon/4$ throughout the disk. Along each disk radius, $\partial_W \phi = -1/\sin\phi$ gives branch-coordinate motion $O(c\sqrt{\varepsilon})$; compact roots move by $O(c\varepsilon)$. The finite continuation cover

and the regular canceled expression of Lemma 7.2 give a uniform complete-pair change $O(\sqrt{\varepsilon})$. Reduce the final ε_0 to make it smaller than both reserved margins $q' - q_1$ and $\eta/2$. Thus same pairs are at most q', cross pairs at most $1 + \eta$, and named pairs are uniformly bounded. The remaining-pair count gives the estimate with $\kappa = -\log(q')/12$, or a smaller common exponent. The bounds apply on the K and current supports, which exclude the upper inverse branch. □

Lemma 7.4 (Branch geometry)**.** *On a sufficiently small original lower-branch arc,*

$$1 - W = a_B u - \mathrm{i} b_B \varepsilon + O(u^2 + \varepsilon|u| + \varepsilon^2), \quad a_B = \sin\theta_B > 0, \quad b_B = 2\sin\theta_* - c_0 a_B > 0.$$

With $\phi = \arccos W$ on the fourth-quadrant sheet,

$$|\phi'(u)| \asymp (|u| + \varepsilon)^{-1/2}, \qquad \int_B |\phi'(u)|\,\mathrm{d}u \le C,$$
$$-\Im\phi(u) \ge c\sqrt{|u| + \varepsilon} \quad (u \le 0), \qquad x'(u) \asymp (u + \varepsilon)^{-1/2} \quad (u \ge 0),$$

where $x = \Re\phi$. For $u \ge C_0\varepsilon$, $-x''(u) \ge cu^{-3/2}$. If $M/\varepsilon \to \infty$ uniformly and $0 \le m \le M/2$,

$$x'(m) - x'(M) \ge c(m + \varepsilon)^{-1/2}, \qquad \frac{|\phi'(m)\phi'(M)|}{x'(m) - x'(M)} \le CM^{-1/2}.$$

If $M/2 < m < M$ and $m \gg \varepsilon$, the latter ratio is at most $C\sqrt{M}/(M - m)$.

On a true-branch plateau of the current put $t = \lambda P/N$, $E = \varepsilon + t$. Uniformly for $0 \le \lambda \le \lambda_*$ and $|u| \le 2r_0$,

$$|\phi_u| \asymp (|u| + E)^{-1/2},$$
$$-\Im\phi \ge c\sqrt{|u| + E} \quad (u \le 0), \qquad \Re\phi_u \ge \tfrac{2}{3}|\phi_u| \quad (u \ge 0).$$

The constants may depend on the fixed τ, but not on $N, \varepsilon, \lambda, P/N$. Here the margins and then ε_0 are small enough, and P is fixed during a plateau variation.

Proof. The square-root normal form controls both sides of the branch. On the negative side it gives attenuation; on the positive side it gives the slope and curvature needed for real coarea. We then repeat the calculation on a deformed branch plateau.

The displayed normal form follows by Taylor expansion of S and the fixed-radius y. Put $D = 1 - W$. The exact factorization $\phi = \sqrt{2D}\,h(D)$, $h(0) = 1$, with analytic h, yields the first estimates and an integrable branch length. On $u \ge C_0\varepsilon$, the leading terms $x'(u) = \sqrt{a_B/2}\,u^{-1/2}(1 + O(\varepsilon/u + u))$ and $-x''(u) \asymp u^{-3/2}$ have the stated signs after increasing C_0 and shortening the arc. If $m \ge C_0\varepsilon$ and $m \le M/2$, the relative error in x' can be made smaller than a fixed fraction of $1 - 1/\sqrt{2}$; this gives the strong difference estimate. If $m < C_0\varepsilon$, its slope is at least $c\varepsilon^{-1/2}$, whereas the slope at M is $o(\varepsilon^{-1/2})$. For comparable m, M, integrate the bound on $-x''$. The sector containing ϕ' also has width less than π; integrating in u gives $|\phi(u) - \phi(v)| \ge c|u - v|$.

The deformed plateau. For the current, write $y = e^{v + \mathrm{i}\vartheta}$, $v = -c_0\varepsilon + \tau t/2$, $\vartheta = -\theta_B + u$. Put $Q = \varepsilon + \tau t$, comparable to E for fixed $\tau > 0$. The exact real and imaginary identities in Lemma 7.3, at the radial center, give

$$D = 1 - W = a_B u - \mathrm{i}\beta + R, \qquad cQ \le \beta \le CQ,$$
$$R \in \mathbb{R}, \qquad |R| \le C(u^2 + Q^2), \qquad D_u = a_B + O(|u| + Q).$$

Positivity of β uses $b_B > 0$ and the positive τt coefficient. The constants in these Q-bounds can be chosen before shrinking τ. The model $a_B u - \mathrm{i}\beta$ has modulus comparable to $|u| + Q$, and its relative perturbation is $O(r_0 + Q)$. Its square root for $u \le 0$ has argument in $[-\pi/2, -\pi/4]$; its inverse square root for $u \ge 0$ has argument in $[0, \pi/4]$. Using the exact factorization of $\phi = \sqrt{2D}\, h(D)$ and its derivative gives the cone and slope bounds after shrinking the fixed margins and ε_0. The remainder allows the $O(E^2)$ real displacement of the branch center, including a positive $\Re D$ at $u = 0$. In particular, since all $|z_i| \le 1$, for a negative branch coordinate

$$\frac{|\phi_v'|}{|1 - Z|} \le \frac{|\phi_v'|}{1 - |Z|} \le \frac{C}{|u_v| + \varepsilon + \lambda P/N}.$$

□

Proposition 7.5 (Nonresonant microcore)**.** *For $N_0 + 2 \le N < D\sqrt{H}$, put*

$$b_N = c_* e^{-(A+4)N}/N, \qquad \psi_N = \prod_i \chi(u_i/b_N),$$

where $\chi = 1$ on $[-1/2, 1/2]$, supported on $[-1, 1]$, and $c_* > 0$ is fixed and small. The all-B microcore contribution obeys, for every $j \le k$,

$$|\partial_s^j I_{\mathrm{micro},N}| \le \exp(C_j N^2)\, b_N^{(N^2-1)/2}.$$

Proof. Nonresonance keeps the angular product away from one on this exponentially small core. A one-phase transport field leaves the root-product kernel simple, and the Vandermonde then supplies the strong particle-number decay.

Uniformly in this window, $N\varepsilon e^{AN} \to 0$ and $\varepsilon/b_N^m \to 0$ for every fixed m. Thus on the microcore and all its cutoff jets,

$$|Y - \xi^N| \le CN(\varepsilon + b_N) \le \tfrac{1}{2} e^{-AN}, \qquad |1 - Y| \ge \tfrac{1}{2} e^{-AN}.$$

The one-phase field $V_i = a_i = S'/W_{u_i}$ freezes each regular ϕ_i, hence Z. In these coordinates the numerator is analytic times $\Delta(\phi)^2$. Repeated Lie differentiation leaves the latter polynomial unchanged and leaves the Z-kernel simple. The Y-power is at most $j + 1$; its gap and the b_N^{-j} cutoff cost are $\exp(O_j(N))$. Analytic products have bound $\exp(O_j(N^2))$.

The branch image lies in a fixed cone of width less than π and has radius $C\sqrt{b_N}$. Decreasing c_* ensures $N\sqrt{b_N}$ is small, so $|1 - Z| \ge c \sum_i |\phi_i|$. The square-root normal form gives, directly on both sides of zero, $|\phi| \ge c\sqrt{|u| + \varepsilon}$ and $|\phi'| \le C(|u| + \varepsilon)^{-1/2}$. Substituting $w = \sqrt{|u| + \varepsilon}$ therefore gives

$$\int_{\mathrm{core}} e^{-t|\phi|} |\,\mathrm{d}\phi| \le 4C \int_{\sqrt{\varepsilon}}^{\sqrt{b_N + \varepsilon}} e^{-ctw}\,\mathrm{d}w \le C \min(\sqrt{b_N}, t^{-1}).$$

Using $1/\sum |\phi_i| = \int_0^\infty e^{-t\sum |\phi_i|}\,\mathrm{d}t$ therefore costs $C^N b_N^{(N-1)/2}$. The Vandermonde costs at most $C^{N(N-1)} b_N^{N(N-1)/2}$. The logarithm of the displayed majorant is

$$-\frac{A+4}{2} N^3 - \frac{1}{2} N^2 \log N + O_j(N^2).$$

In particular it is at most $-(A + 4)N^3/2 + O_j(N^2)$. Thus the sum of these bounds over all $N \ge N_0 + 2$ is finite. □

Proposition 7.6 (The all-branch complement)**.** *The remaining original all-B contribution, for $N_0+2 \le N < D\sqrt{H}$, is bounded through every $j \le k$ by*

$$\exp(C_j N \log(N+1))(H+N)^{C_j}.$$

Proof. Telescope $1-\psi_N$; each term has a named anchor a with $|u_a| \ge b_N/2$, including all cutoff jets. Let $t = N^{-1}\sum u_i$, $\rho^2 = \sum(u_i-t)^2$, and split smoothly at $\rho_N = e^{-BN}$, using a function of ρ^2/ρ_N^2. Put

$$\Gamma_* = \sup_{\substack{n\in\mathbb{N}\\ n\ge N_0+2}} \frac{\log(16n/c_*)}{n} < \infty.$$

Choose $B > A + 4 + \max\{1, \Gamma_*\}$, increasing it as needed after the finite-jet constants have been fixed. Then

$$2\rho_N \le b_N/8 \qquad \text{for every } N \ge N_0+2.$$

Far from full equality. On the far support $\rho \ge \rho_N$, the maximal separation is at least $\rho_N/\sqrt{N}$. The rational two-phase Lie calculus costs $\exp(O_j(N\log N))$, including every exponential cutoff jet. The undifferentiated pairs retain $\exp(-\kappa N^2 + O_j(N))$. It remains to integrate two simple global kernels against $\prod_i |\phi_i'|\,\mathrm{d}u_i$.

If all coordinates are nonnegative, their maximum $M \ge b_N/2$. Select the actual minimum m and maximum M only now, after differentiation. For fixed other variables the real phase map is

$$(m, M) \longmapsto (F, G) = (m + M + \text{const},\ x(m) + x(M) + \text{const}).$$

Its Jacobian $x'(m) - x'(M) > 0$ makes it injective on each fixed-F extreme-index cell. Lemma 7.4 bounds the ratio of its two branch measures to that Jacobian by $\exp(O(N))$: use $M \ge b_N/2$ for $m \le M/2$, and $M - m \ge \rho_N/\sqrt{N}$ otherwise. Only m, M vary here; all other angles are fixed. Each unwrapped phase-image interval has bounded length, independent of N, since m, M and $x(m), x(M)$ range over fixed bounded intervals. The additive constants translate these intervals without changing their lengths. Thus each simple periodic phase integral costs $O(H+N)$. The other branch lengths cost C^N.

For negative coordinates choose $\delta_N = e^{-C_1 N}$ with $C_1 > A+4+\max\{1,\Gamma_*\}$. Then $\delta_N \le b_N/16$ for every $N \ge N_0+2$. If some negative $|u_v| \ge \delta_N$, then $|1-Z| \ge e^{-O(N)}$; integrate the remaining Y-kernel in the named anchor, whose measure is at most $e^{O(N)}\,\mathrm{d}u_a$. If all negative magnitudes are smaller than δ_N, the named anchor is positive. Choose one negative v; the bound

$$\frac{|\phi'(u_v)|}{|1-Z|} \le \frac{C}{|u_v| + \varepsilon}$$

has integral $O(H+N)$. The positive anchor integrates the Y-kernel with cost $e^{O(N)}(H+N)$. The scale inclusion already covers every finite order; any remaining finite-order constants only enlarge the numerical bound. This proves the far bound $\exp(-\kappa N^2 + O_j(N\log N))(H+N)^2$.

On the near support, $|u_i - t| \le \rho \le 2\rho_N \le b_N/8$. The anchor $|u_a| \ge b_N/2$ gives $|t| \ge 3b_N/8$, so all coordinates have the same sign and $|u_i| \ge b_N/4$. The pointwise branch-measure product has the bound

$$\prod_i |\phi_i'| \le (C b_N^{-1/2})^N = \exp\left(\frac{A+4}{2}N^2 + \frac{N}{2}\log(N/c_*) + O(N)\right).$$

Segments joining two coordinates stay on this same-sign support, so each collision difference is at most $Cb_N^{-1/2}\rho$. The collision coefficient and the branch measures together are therefore bounded by $C^{N^2}b_N^{-N^2/2}$, whose logarithm is

$$\frac{A+4}{2}N^3+\frac{N^2}{2}\log(N/c_*)+O(N^2).$$

Fixed-j coefficient and derivative costs can be included in the lower-order $O_j(N^2)$ term. Its constant is fixed before B. These bounds are determined on the B-independent anchor neighborhood; the positive cubic coefficient is independent of B. Lie and cutoff jets lose at most $2j+1$ powers of ρ. The shape measure contributes ρ^{N-2}. For negative t, the Z-gap is $e^{-O(N)}$ and one mean phase integral suffices. For positive t, strict concavity and zero first shape derivative give $-\partial_\rho\Re\sum\phi_i\geq c\rho$, while the Y-phase is $Nt+\text{const}$. Their real Jacobian is at least $cN\rho$. After coarea the conservative remaining power is

$$L=N^2-2j-4\geq N^2-2k-4\geq 4N_0+2>0.$$

Bound this power by $(2\rho_N)^L$ before integrating the phase kernels. At fixed shape direction, $F=Nt+\text{const}$ fixes t, and strict monotonicity of G in $\rho>0$ gives multiplicity one for the unwrapped map. Each phase lies in an interval of length at most CN: the mean angle ranges over a fixed interval and each individual branch phase is bounded. The periodic-kernel estimate in Appendix D therefore costs at most $CN^2(H+N)^2$ for the two phases. For negative t, the sole mean phase costs at most $CN(H+N)$, since the Z-kernel is separated. Absorb these factors in $e^{C_jN\log(N+1)}$. With the remaining shape power bounded on the support, the estimate is

$$\exp\left(\frac{A+4}{2}N^3+\frac{N^2}{2}\log N+C_jN^2\right)(2\rho_N)^L(H+N)^2.$$

Its logarithm, apart from the simple-kernel factor, is at most $((A+4)/2-B)N^3+\frac{1}{2}N^2\log N+O_j(N^2)+B(2j+4)N$. The already required $B>A+4$ makes this tend to minus infinity. Finitely many remaining N's cost a fixed, possibly very large, ε-independent constant. At fixed $\varepsilon>0$ the same positive collision power makes the full-equality puncture flux vanish at every preceding Lie stage. Thus the identities used on the punctured charts have no residual boundary term. The near and far bounds prove the proposition. □

8 Compact and mixed sectors; Stokes estimates

The remaining intermediate sectors contain a compact coordinate. A left root gives a modulus gap; a mixed compact/branch pair supplies a stable two-phase field; and the all-right sector uses strict phase curvature. The large current parameter is handled separately on a protected complex disk. These estimates complete the targets stated in Section 5.

Proposition 8.1 (Separated mixed and left sectors)**.** *For $N_0+2\leq N<D\sqrt{H}$, the non-all-B original K sectors with a compact left root or a separated compact/branch pair, and the corresponding small-λ Stokes sectors, obey* (5.7)*. The differentiated current density obeys the per-order bound uniformly for $0\leq\lambda\leq\lambda_*$.*

Proof. Use hybrid coordinates consisting of regular ϕ_i for true branch indices J and original angles for the compact indices. All coefficient jets are bounded in these coordinates. For Stokes the named q lies on $p = 1, m = 0$, and every true branch lies on $m = 1, p = 0$; their angle variations preserve both $P = \sum p_i$ and $\sum m_i$ exactly. The original K has zero deformation.

Put $a_i = (\partial_s \phi_i)/b_i$, $b_i = \partial_{\theta_i} \phi_i$, $A = \sum_{i \in J} a_i$, and $D = \sum_{i \notin J} \partial_s \phi_i$. With $j = \min J$ fixed for the labelled term, let

$$X = \frac{D + b_q A}{b_j - b_q}, \qquad V_i = a_i \ (i \in J \setminus \{j\}), \quad V_j = a_j + X, \quad V_q = -A - X.$$

All other velocities vanish. Direct substitution freezes Y, Z and every unselected branch phase. The residual branch velocity is

$$(\partial_s - V \cdot \nabla)\phi_j = -\frac{D + b_q A}{1 - b_q/b_j}.$$

The support choice gives $|b_q/b_j| \le 1/4$. Since $1/b_j$ is analytic and vanishes in its regular branch coordinate, this residual, its finite jets, and the compact velocity have only polynomial N-cost, uniformly in ε, λ. Here the hybrid top form can be written explicitly. Up to the unchanged factor $(Z^{-1} + Y^{-1})/[N!(1 - Z)(1 - Y)] \prod_{i<l} P_{il}$, its measure is

$$\prod_{i \in J} \left[-\frac{z_i}{\pi(1 - y_i^{-2})} \, \mathrm{d}\phi_i \right] \prod_{i \notin J} \left[\frac{R_i y_i d_i}{2\pi \mathrm{i}} \, \mathrm{d}\theta_i \right].$$

Indeed $d_i = \mathrm{i}$ on every true branch plateau, the angular determinant is $\prod d_i$, and the hybrid change of variables has triangular determinant $\prod_{i \in J} b_i$. This retains the full Jacobian of the coupled contour.

For Stokes, $V \cdot \nabla P = V \cdot \nabla \sum m_i = 0$ identically on an open neighborhood of the support, and these identities persist under parameter differentiation. The current multiplier is $-2\mathrm{i}\tau \, \partial_q \psi$; its finite derivatives, and the derivatives of the nested partition, stay on these plateaus. Use the pullback recurrence (D.4) in the hybrid coordinates. Its residual coefficients have no difference pole, because $1 - b_q/b_j$ is separated from zero. Each fixed operation changes finitely many analytic pair or one-body factors and introduces at most polynomial N-cost. The remaining pair factors therefore retain $C_j^N N^{C_j} e^{-\kappa N^2}$.

For positive branch coordinates, Lemma 7.4, including its current version, gives $\Re b_j \ge (2/3)|b_j|$. The compact slope is dominated by the true branch slope; for a Stokes upper-right q it has the opposite sign in the limit. The phase map $(\theta_q, u_j) \mapsto (\sum \theta_i, \Re \sum \phi_i)$ has Jacobian $\Re(b_j - b_q) \ge (2/3 - 1/4)|b_j|$ and is injective on fixed first-phase slices. It absorbs the selected branch measure, and each remaining phase kernel integrates with cost $O(H + \log N)$. Every unselected branch measure has bounded integral.

If a branch coordinate $u_v < 0$, its attenuation and branch measure give instead

$$\frac{|\phi_v'|}{|1 - Z|} \le \frac{C}{|u_v| + \varepsilon + \lambda P/N}.$$

Integrate this factor in u_v, and the Y-phase in the compact q. Both cost at most $C(H + \log N)$. The selected integrations take place on plateaus of the occupancy functions, with P fixed. The sign split is made in the final absolute integral.

If a compact left root occurs, Z has a fixed modulus gap. Set $V_i = a_i$ for branches and $V_q = -A$, preserving Y and freezing every branch root. Differentiating the separated Z-kernel

costs only polynomial N, and q integrates the simple Y-kernel. A left q itself is allowed. The fixed support partition exhausts the asserted cases. □

Proposition 8.2 (Compact right sectors)**.** *The all-compact right original K and small-λ Stokes sectors satisfy the same per-order bound in* (5.7)*.*

Proof. With all roots in the compact right region, strict curvature replaces the branch slope separation. The rational atlas resolves pair coincidences while retaining the full-collision vanishing needed to close the integration-by-parts boundary terms.

On a fixed compact right interval R, separated from both branch points, put $g(\theta) = \arccos(S_* - \cos\theta)$. Direct differentiation gives

$$g''(\theta) = -\frac{S_*(1 - \cos\theta\cos g(\theta))}{\sin^3 g(\theta)} \le -c_R < 0.$$

The coupled contour is smooth in the angles and analytic in s. Its fixed finite angular jets differ from their limiting values by at most $C_j N^{C_j}(\varepsilon + \lambda)$. Indeed $P/N \in [0,1]$, each fixed angular jet of P/N is uniformly bounded, and every deformation term carries a factor λ. The undeformed parameter differs from s_* by $O(\varepsilon)$. The root map has bounded jets on the fixed regular compact neighborhood; the chain and product rules and the sum over indices cost only a polynomial in N for fixed j. For the Hessian, passing from entrywise bounds to operator norm costs at most one further factor N. Since $N \le D\sqrt{H}$, $\varepsilon = e^{-H}$, and $0 \le \lambda \le e^{-\alpha_0 H}$, these errors tend to zero uniformly after all jet constants and $D, \alpha_0 > 0$ are fixed. In particular $\nabla^2 G \le -(c_R/2)I$ for $G = \Re \sum \phi_i$.

Write $f_i = \partial_{\theta_i} \log Y$, $g_i = \partial_{\theta_i} \sum \phi_i$, and $Q_s = \partial_s \sum \phi_i$. Permutation symmetry makes $D_{pq} = f_p g_q - f_q g_p$ zero at $\theta_p = \theta_q$. Division by the difference is justified for these smooth angular functions by the integral of their transverse derivative; no analytic extension of the deformation bumps is assumed. At the limit it equals $\mathrm{i}[g'(\theta_q) - g'(\theta_p)]$. The strict curvature and the finite-jet perturbation bound yield

$$D_{pq} = (\theta_p - \theta_q) C_{pq}, \qquad |C_{pq}| \ge c_R/2.$$

The field $V_p = -Q_s f_q / D_{pq}$, $V_q = Q_s f_p / D_{pq}$ freezes Y, Z. Its simple difference pole, divergence, and rational real-angle weights have the two-power-per-Lie-order bound of Appendix D.

For separation $d \ge N^{-K_0}$, all selected losses are polynomial in N, and the unchanged pair factors retain quadratic suppression. For $d < N^{-K_0}$, use $\theta_i = t + \rho\omega_i$, with $\sum \omega_i = 0$, $\sum \omega_i^2 = 1$. Permutation symmetry gives $\partial_\rho G(t, 0, \omega) = 0$, so $-\partial_\rho G \ge c_R \rho/2$. The Y-phase is exactly Nt; its real radial shifts do not alter that phase. The coarea Jacobian in t, ρ is at least $c_R N \rho/2$. Convexity of the interval cube gives one interval on each shape ray, and monotonicity gives multiplicity one there.

At a collision $\theta_i = \theta_j$ the symmetric coupled shifts are equal, so $y_i - y_j$ vanishes to first order. The raw numerator is bounded by $C^{N^2} \rho^{N(N-1)}$, including its regular pair denominators. After j Lie jets and one conservative flux/cutoff loss, its remaining coarea power is $L = N^2 - 2j - 4 \ge 4N_0 + 2 > 0$. The inequalities $d \le 2\rho$ and $\rho \le \sqrt{N}\, d$ give $\rho \le \sqrt{N}\, N^{-K_0}$ in the near region. Taking fixed $K_0 > 1$ sufficiently large gives

$$C^{N^2} (\sqrt{N}\, N^{-K_0})^L e^{O_j(N \log(N+1))} (H + \log N)^2 \le e^{C_j N \log(N+1)} (H + N)^{C_j}.$$

The simple phase kernels have attenuation floors $cN\varepsilon$ in this window. On each fixed shape ray, $F = Nt$ and bounded individual phases put the unwrapped image inside a rectangle with side

lengths at most CN. The multiplicity-one statement above, followed by the interval estimate in Appendix D, bounds its phase integral by $CN^2(H+\log N)^2$. The $O(N^2)$ translated period boxes give the same factor, absorbed in $e^{O_j(N\log(N+1))}$ in the displayed estimate. On the separated region the coarea $1/\rho$ costs at most a polynomial. Selected-diagonal fluxes vanish by the rational numerator, and full-equality fluxes vanish by the positive collision exponent at fixed ε. The actual Stokes multiplier is smooth with bounded finite jets and its q remains on a plateau, uniformly for $\lambda \le \lambda_*$. □

Lemma 8.3 (Protected large-current disks)**.** *On a named Stokes support, let $\lambda_* \le \lambda \le 1$. For a sufficiently small constant $c > 0$, depending only on the fixed point and previously chosen margins, the density is holomorphic on*

$$|s - s_\varepsilon| \le \delta_s, \qquad \delta_s = c\lambda_*/N^2.$$

On this disk its product gaps are at least $c'\lambda$, its complete pairs have bound $C^N e^{-\kappa' N^2}$, and its one-body factors have integrable majorants with total cost C^N.

Proof. For the large current parameter, the named upper index supplies damping that remains after a small complex perturbation. We use this margin to choose the disk, allow controlled compact-root inflation, and retain both product gaps.

There is a named upper compact q with $p_q = 1$, so $P \ge 1$. Fix a lower branch neighborhood inside $m = 1, p = 0$ and put $a_B = \min|\sin\theta| > 0$ there. With $t = \lambda P/N$, the exact imaginary-part formula gives

$$\Im W(s_\varepsilon) \ge b_0\varepsilon + \tfrac{1}{2}\tau a_B t \ge b_\tau(\varepsilon + t), \qquad b_\tau = \min(b_0, \tau a_B/2) > 0.$$

Indeed the increase from the outward shift is at least $\tau a_B t/2$, since $\cosh v \ge 1$ for real v. Write the disk constant as c_P. After fixing τ, require

$$L_S c_P \le b_\tau/4.$$

Since $t \ge \lambda_*/N$, throughout the disk

$$|\Delta W| \le L_S c_P \lambda_*/N^2 \le b_\tau t/(4N) \le b_\tau(\varepsilon + t)/4.$$

Thus these branch-near roots remain on the lower interior sheet.

Every non-plateau support angle has a fixed limiting distance $d_1 > 0$ from $W = 1$. Compute it on the preliminary endpoint neighborhoods and the complement of the fixed lower plateau, before shrinking δ, τ; the upper branch is excluded. All limiting roots also have distance at least S_* from $W = -1$. For $d = \min(d_1, S_*) > 0$, the choices of Lemma 7.2, followed by a reduction of c_P, ensure

$$\operatorname{dist}(W, \{1, -1\}) \ge d - C(\varepsilon + \tau + \delta) - L_S\delta_s \ge d/2.$$

Here δ is the fixed endpoint width. Use the compact analytic continuations specified in that lemma. Their motion is $O(\delta_s)$; their modulus can exceed one. The branch-coordinate motion is at most

$$C_\tau \frac{\delta_s}{\sqrt{\varepsilon + t}} \le C_\tau c_P \frac{\sqrt{\lambda_*}}{N^{3/2}}.$$

Reduce c_P further so these motions consume less than the reserved $q' - q_1$ and $\eta/2$ pair margins and retain the compact denominator margins. The regular double-branch expression controls the

remaining neighborhood. This proves same-group bound q', cross bound $1+\eta$, and the named bound throughout the disk. The direct count in Lemma 7.2 gives the same quadratic exponent, including deletion of at most j pairs.

At the radial center the named upper root has $|z_q| \le e^{-c_q\lambda}$, $c_q > 0$, because its inward shift is $2\tau\lambda$ and its arc has positive sine. Branch roots remain interior on the disk and compact roots obey $|z_i(s)| \le |z_i(s_\varepsilon)|e^{C\delta_s}$. Adding $Cc_P \le c_q/2$ to the disk restrictions gives

$$|Z| \le e^{-c_q\lambda + CN\delta_s} \le e^{-c_q\lambda/2}.$$

Also $|Y| \le e^{-3\tau\lambda P/2}$, independently of s. Both product gaps are therefore at least $c'\lambda$. Choose c_P to meet these branch and product margins after fixing τ and before choosing the final ε_0.

Finally put $t = \lambda P/N$. The real displacement of the branch center is $O(\varepsilon^2 + t^2 + \delta_s)$, and the imaginary margin is at least $c_2(\varepsilon + t)$. Split into $|u| \le C(\varepsilon + t)$ and its complement. The imaginary margin bounds the square root in the first case, and the linear real term bounds it in the second, with the earlier choices of τ and then c_P sufficiently small. Hence

$$|R(y; s)| \le C(|u| + \varepsilon)^{-1/2}.$$

This is a majorant independent of the coupled scalar P; no integration treats P as a free constant. Compact one-body factors are bounded. The inverse-product numerator, the exact Jacobian, and its Stokes replacement minor cost C^N. These facts prove all the stated assertions. □

Proposition 8.4 (Large-λ current)**.** *For $N < C_H(H+1)^2$, the differentiated current integrated over $\lambda_* \le \lambda \le 1$ satisfies* (5.5).

Proof. Apply Cauchy to the protected current slice, sum its Gaussian particle-number envelope, and integrate the remaining current-parameter power. The resulting stronger estimate implies the target budget.

Lemma 8.3 bounds the absolute integral of the undifferentiated density on its common disk by $C^N e^{-\kappa N^2}\lambda^{-2}$, including the named-index count in C^N. The majorant is integrable and uniform on that disk, so the integrated density is holomorphic there. Cauchy's inequality of order j adds $C_j N^{2j}\lambda_*^{-j}$. Integration in λ costs at most λ_*^{-1}. Summing the convergent series $\sum_N C^N N^{2j} e^{-\kappa N^2}$ gives even the stronger bound $C_j\lambda_*^{-j-1}$. Its weaker version $C_j\lambda_*^{-j-2}$ is (5.5). By (5.1), this is $o(\varepsilon^{-1/2})$. □

Proposition 8.5 (The remaining high orders)**.** *The bound* (5.6) *holds on $D\sqrt{H} \le N < C_H(H+1)^2$.*

Proof. Use Lemma 7.3, which establishes the original $c\varepsilon$ disk, the $c\varepsilon$ product gaps, integrable one-body majorants, and complete pair bound $C^N e^{-\kappa N^2}$. Cauchy's inequality of order j gives $C_j^N \varepsilon^{-j-2} e^{-\kappa N^2}$. Integrating the small-$\lambda$ current only reduces this bound. The product gap used here is $c\varepsilon$, uniform for unbounded N. At fixed ε, a lower bound $1 - r^N \ge cN\varepsilon$ would eventually exceed one. □

9 Summation and the natural boundary

Theorem 9.1 (Uniform differentiated tail)**.** *For each $0 \le j \le k$,*

$$\sum_{\substack{N > N_0 \\ N \text{ even}}} |T_N^{(j)}(s_\varepsilon)| = o(\varepsilon^{-1/2}).$$

Proof. Appendix C gives local normal convergence of each fixed differentiated series. Split the sum of these derivatives at the ε-dependent thresholds. The ultra-high tail is negligible by Proposition 5.2. For the remaining orders use the exact identity (5.2). The F-sum is $\exp(O_j(\log^2 H))$. The large-λ current is at most $C_j \exp(\alpha_0(j+2)H) = o(e^{H/2})$.

Take a common positive κ smaller than all pair-contraction constants, and then choose D with $\kappa D^2 > k+2$. The high-N original and small-λ pieces have sum at most

$$\exp((j+2-\kappa D^2)H + O_j(\sqrt{H})) = o(e^{H/2}).$$

All intermediate pieces are covered by the fixed partition, Propositions 7.5, 7.6, 8.1, and 8.2. Their sum has logarithm at most $O_j(\sqrt{H}\log(H+2)) = o(H)$. The small-$\lambda$ integration contributes at most its length $\lambda_* \le 1$. The same choices work for all $j \le k$, since this is a finite set. This proves the claimed absolute tail estimate. □

Theorem 9.2 (Bulk noncancellation criterion)**.** *Assume the first-form-factor result of Theorem 4.4 and the absolute tail estimate of Theorem 9.1 at every selected point. Then $\mathbb{T}$ is a natural boundary of the exterior bulk susceptibility germ.*

Proof. Write $U = \sum_{N \text{ even}} T_N$. The finitely many terms below N_0 have bounded k-th derivatives. The first term has $T_{N_0}^{(k)} = 2L_\beta \varepsilon^{-1/2} + o(\varepsilon^{-1/2})$. The absolute higher-tail estimate excludes cancellation, giving

$$U^{(k)}(s_\varepsilon) = 2L_\beta \varepsilon^{-1/2} + o(\varepsilon^{-1/2}).$$

For $j < k$, the finite first and lower terms are bounded and the higher tails are $o(\varepsilon^{-1/2})$. Since $\mathcal{M}^2$ is analytic and nonzero near s_*, Leibniz's rule in (2.2) therefore gives

$$\mathcal{X}^{(k)}(s_\varepsilon) = 4\mathcal{M}^2(s_*)L_\beta \varepsilon^{-1/2} + o(\varepsilon^{-1/2}), \qquad \mathcal{M}^2(s_*)L_\beta \neq 0.$$

Holomorphic extension to a neighborhood of s_* would bound this derivative on a sufficiently short radial segment, a contradiction. The selected points are dense in the upper-right arc; evenness and conjugation give density in all four quadrants. A neighborhood of any point of $\mathbb{T}$, including an axis point, contains such a selected point. Hence no neighborhood permits an exterior holomorphic extension. The locally nonzero analytic factor $\beta_{\text{phys}}(s)$ transfers the same conclusion to the physical susceptibility on its chosen continuation. □

Corollary 9.3. *For the zero-field isotropic infinite square-lattice Ising model, the unit circle in $s = \sinh(2\beta_{\text{phys}}J)$ is a natural boundary of the bulk susceptibility germ continued from the low-temperature side.*

Proof. Apply Theorem 9.2 using Theorems 4.4 and 9.1. □

Corollary 9.4 (No smooth extension on an arc)**.** *The exterior normalized bulk germ admits no real C^∞ extension through any nonempty open unit-circle arc. More precisely, for an arc of angular length $0 < \ell \le 2\pi$, there is a point on it and a finite integer k at which no real C^k extension agreeing with the exterior germ on a neighborhood exists. One may take $k = 2p^2 - 1$ for any prime $p \ge 11$ satisfying $2\sqrt{10}\pi/p < \ell$. In particular an elementary sufficient bound is*

$$k < \max\{968,\, 320\pi^2\ell^{-2}\}.$$

The same assertion holds for each chosen physical continuation.

Proof. Put $h = 2\pi/p$ and $m = \lfloor (p-1)/4 \rfloor$. The selected family contains the points with adjacent angles $jh, (j+1)h$, $1 \le j < m$; write their arguments as $t_j = \arccos((\cos(jh) + \cos((j+1)h))/2)$. They are strictly ordered and lie between these two grid angles. Concavity of cosine gives $t_j \ge (j+1/2)h$, so consecutive gaps are less than $3h/2$. At the real axis, convexity of $u \mapsto \cos\sqrt{u}$ on $[0, \pi^2]$ gives $t_1 \le \sqrt{5/2}\, h$. Its derivative test is $\sin v - v\cos v \ge 0$ for $0 \le v \le \pi$. At the other end, $\pi/2 - t_{m-1} \le (p/4 - m + 1/2)h \le 5h/4$. Include sign and conjugation images. Every circular gap in this finite family is at most $\sqrt{10}\, h = 2\sqrt{10}\pi/p$, including both sorts of axis gap. Thus every open arc longer than this bound contains a selected image point, with derivative order $k = 2p^2 - 1$. Bertrand's prime bound applied to $x = \max\{11, 2\sqrt{10}\pi/\ell\}$ supplies $x < p < 2x$ and the displayed elementary bound. All point-dependent constants are fixed after this finite choice of p, a, b.

At the selected point Theorem 9.2 gives a nonzero inverse-square-root singularity of the k-th derivative. For a real C^k extension F, the curve $\varepsilon \mapsto F((1+\varepsilon)s_*)$ is C^k through zero. On its exterior segment its k-th real derivative is $s_*^k \mathcal{X}^{(k)}((1+\varepsilon)s_*)$, which is unbounded, a contradiction. Symmetry transfers this obstruction to the chosen image point. A nonzero local analytic physical prefactor preserves the highest singular coefficient by the same product rule.

As $\ell \downarrow 0$, the prime number theorem permits the choice $p = (1 + o(1))2\sqrt{10}\pi/\ell$, giving $k \le (80\pi^2 + o(1))\ell^{-2}$. For a fixed arc the finite-prime bound above gives the sufficient order for an extension agreeing with the exterior germ. □

A Cyclotomic field intersection

This appendix supplies the field-intersection fact used to exclude smaller even Nickel orders. Its proof also makes explicit the minimal polynomial input behind the sparse cosine relation.

Lemma A.1. *If p is prime and $p \nmid n$, then $\mathbb{Q}(\zeta_p) \cap \mathbb{Q}(\zeta_n) = \mathbb{Q}$.*

Proof. We prove the degree statement by the elementary cyclotomic irreducibility argument and then compare the two field degrees. The coprimality assumption is used in this comparison.

We recall the needed elementary irreducibility argument. The cyclotomic polynomial Φ_m is monic with integer coefficients, obtained successively from $X^m - 1 = \prod_{d|m} \Phi_d$. Let f be the monic irreducible polynomial of a primitive m-th root ζ. By Gauss's lemma, $f \in \mathbb{Z}[X]$ divides Φ_m. For a prime $q \nmid m$, suppose ζ^q belonged to a different irreducible factor g of Φ_m. Then $f \mid g(X^q)$. Modulo q, $g(X^q) = g(X)^q$; hence the reductions of f and g share a factor. Their product divides $X^m - 1$ modulo q, making that polynomial have a repeated factor. But its derivative is mX^{m-1}, coprime to $X^m - 1$ in characteristic q, a contradiction. Thus taking a q-th power preserves the conjugacy class for every prime $q \nmid m$. Factoring any positive integer coprime to m into such primes shows that every primitive m-th root is a conjugate. Therefore Φ_m is irreducible and $[\mathbb{Q}(\zeta_m) : \mathbb{Q}] = \varphi(m)$.

For $p \nmid n$, the compositum $\mathbb{Q}(\zeta_p, \zeta_n) = \mathbb{Q}(\zeta_{pn})$, by Bezout's identity. Its degree is $\varphi(pn) = \varphi(p)\varphi(n)$. This is the maximal possible product degree. If the intersection contained a field of degree greater than one, a basis calculation over that intersection would make the compositum degree strictly smaller than this product. The intersection is therefore $\mathbb{Q}$. □

For the sparse relation used in Theorem 3.1, a symmetric polynomial representing $c_0 + \sum c_j(\zeta^j + \zeta^{-j})$ has degree at most $p-1$. Its vanishing makes it a scalar multiple of Φ_p. Comparing its paired coefficients gives precisely $c_0 = c_1 = \cdots = c_d$. The requirement $d \ge 5$ is what rules out a nonzero relation involving at most four nonconstant positions.

B The order of local integrations

The order of integration is part of the first-term proof. This appendix explains why the mean residue comes before the shape scaling and then evaluates the same nonzero coefficient in closed form. The nonzero period in Lemma 4.2 supplies the first singular coefficient used in the noncancellation argument. In the homogeneous first-chart scaling $v = \varepsilon w$, $t = \sqrt{\varepsilon}\,\tau$, the two global denominators are

$$\varepsilon(Nc_0 - \mathrm{i}w), \qquad \varepsilon(NA_0 + \mathrm{i}bw - \mathrm{i}dR^2).$$

The first pole is below and the second above the real w-line. For fixed $R^2 = |\tau|^2$,

$$\int_{\mathbb{R}} \frac{\mathrm{d}w}{(Nc_0 - \mathrm{i}w)(NA_0 + \mathrm{i}bw - \mathrm{i}dR^2)^{k+1}} = \frac{2\pi}{(N(A_0 + bc_0) - \mathrm{i}dR^2)^{k+1}}.$$

The combined damping is $N(A_0 + bc_0) = Q$, independent of the arbitrary evaluation contour radius. The shape volume scales as $\varepsilon^{(N-1)/2}$, the Vandermonde as $\varepsilon^{N(N-1)/2}$, and the mean measure as ε. The two denominator powers are ε^{k+2}. The net exponent is $-1/2$.

The high-order Z-ridge in the joint absolute integrand moves with R^2. Take the exact mean residue on the fixed physical rectangle first. The resulting shape integral has the integrable R^{-2} tail needed for the limit.

The positivity of $\mathscr{C}_N$ follows directly: the squared Vandermonde is nonnegative on the real shape sphere and positive in a neighborhood of any zero-sum vector with distinct coordinates. That sphere has finite measure. Thus the beta period's nonvanishing follows from an ordinary positive angular integral and nonzero gamma factors.

Closed form of the same coefficient. The coefficient also has the following closed form. Set $P_N = \prod_{m=1}^{N-1}(m!/2^m)$, $H_{\alpha\beta} = S_*(1 - \cos\alpha\cos\beta) = \sin^2\alpha\cos\beta + \sin^2\beta\cos\alpha > 0$, and $\nu = a_0 - 1 = k - 1/2$. The Gaussian Vandermonde integral is

$$\int_{\mathbb{R}^N} e^{-|x|^2/2}\Delta(x)^2\,\mathrm{d}x = (2\pi)^{N/2}\prod_{m=1}^{N} m!.$$

One direct derivation expands the two Vandermonde determinants in monic Hermite polynomials; orthogonality gives $N!$ times the product of their squared norms $\sqrt{2\pi}\,m!$, $m = 0, \dots, N-1$. These norms follow by repeated integration by parts from $H_m = (-1)^m e^{x^2/2}(\,\mathrm{d}/\,\mathrm{d}x)^m e^{-x^2/2}$. Separate the unit mean coordinate from the zero-sum hyperplane and then use polar coordinates there. Its radial integral is $2^{a_0-1}\Gamma(a_0)$, while the mean integral is $\sqrt{2\pi}$. Remembering the coordinate-volume factor in Lemma 4.2 gives

$$\mathscr{C}_N = \frac{(2\pi)^{(N-1)/2}\prod_{m=1}^{N} m!}{\sqrt{N}\,2^{a_0-1}\Gamma(a_0)}.$$

Substitute this and $k!B(a_0, 1/2) = \sqrt{\pi}\,\Gamma(a_0)$ into the already proved L_β. All powers of $\sin\beta$ cancel, and the result is

$$L_\beta = (-1)^k e^{-\mathrm{i}k\theta_*} e^{\mathrm{i}\pi a_0/2}\frac{(N\sin\theta_*)^\nu P_N}{\pi^{(N-2)/2}\sqrt{N}\,H_{\alpha\beta}^{a_0}}. \tag{B.1}$$

For even N, $a_0 = k + 1/2$ and $\Gamma(a_0) = (2k)!\sqrt{\pi}/(4^k k!)$, so the angular constant as well as L_β is expressible through factorials and elementary factors. This evaluation uses exactly the earlier continuation of $(-\mathrm{i}d)^{-a_0}$.

C Pfaffian identities and exterior convergence

The tail estimates use the Pfaffian identity at finite particle number. The full-sum argument also needs convergence on compact subsets of the exterior before any boundary limit is taken. The two facts are proved here with their distinct domains and constants. We use the classical Schur identity and Hadamard's determinant inequality. The proof below fixes the sign convention in the Schur identity.

Lemma C.1 (Schur identity)**.** *For even N and $A_{ij} = (v_i - v_j)/(1 - v_i v_j)$,*

$$\operatorname{Pf} A = \prod_{i<j} \frac{v_i - v_j}{1 - v_i v_j}.$$

Proof. Induction reduces the rational identity to its residues and its value at a regular point. This also fixes the sign convention used in the Pfaffian estimates.

Set $x_i = (1 + v_i)/(1 - v_i)$, so the matrix entries are $(x_i - x_j)/(x_i + x_j)$. Induct on even N, regarding both sides as rational functions of x_N. At a simple pole $x_N = -x_i$, Pfaffian expansion in the last index gives residue $(-1)^{i+1} 2x_i$ times the $(N-2)$-Pfaffian. On the product side, each index between i and N contributes one minus sign when its two paired factors are evaluated there; the total sign is $(-1)^{N-i-1} = (-1)^{i+1}$. The residues agree by induction. The difference has no finite poles and is bounded at infinity, so is constant. At $x_N = x_{N-1}$, both expressions vanish, fixing that constant to zero. Degenerate cases follow as rational identities by continuation. □

If an even m-by-m antisymmetric matrix has entries at most C, then Hadamard's determinant inequality and $(\operatorname{Pf} A)^2 = \det A$ give $|\operatorname{Pf} A| \le C^{m/2}(m-1)^{m/4}$. The number of perfect matchings is $(N-1)!! = N!/(2^{N/2}(N/2)!)$. These are different bounds; the selected-contour proof uses bounded minors and an explicit expansion only for exceptional compact pairs.

Fix a compact exterior neighborhood K and a slightly larger compact exterior neighborhood K_+ with $K \subset \operatorname{int} K_+$. Choose an admissible radius $r < 1$ and $q < r$ such that every interior root satisfies $|z(y;s)| \le q$ for $s \in K_+$ and $|y| = r$. On this compact set the dispersion quantity is bounded, say $|W(y;s)| \le M_W$. The equation $z + z^{-1} = 2W$ gives

$$|z|^{-1} = |2W - z| \le 2M_W + q, \qquad m := (2M_W + q)^{-1} \le |z|.$$

Thus individual one-body factors and pair matrix entries have bounds independent of N, but the global inverse products need an exponential bound. In fact, $|Y^{-1}| = r^{-N}$, and

$$|Z^{-1} + Y^{-1}| \le m^{-N} + r^{-N} \le 2C_1^N, \qquad C_1 := \max\{m^{-1}, r^{-1}\}.$$

For even $N \ge 2$, the global gaps satisfy $|1 - Z| \ge 1 - q^N \ge 1 - q^2$ and $|1 - Y| \ge 1 - r^N \ge 1 - r^2$. Applying Hadamard's inequality to the two Pfaffians in (2.3), and absorbing the bounded one-body factors and contour measures into a constant to the power N, yields

$$|T_N(s)| \le C_0 \frac{C_2^N N^{N/2}}{N!} \le C_0 \left(\frac{eC_2}{\sqrt{N}}\right)^N, \qquad s \in K_+.$$

Here C_0, C_2 are independent of N, and the last inequality uses $N! \ge (N/e)^N$. Choose $d > 0$ such that each closed d-disk centered in K lies in K_+. Cauchy's inequality gives, for every fixed nonnegative integer j,

$$\sup_{s \in K} |T_N^{(j)}(s)| \le j! d^{-j} C_0 \left(\frac{eC_2}{\sqrt{N}}\right)^N.$$

The majorant is summable over even N and justifies termwise differentiation of every fixed order on compact subsets of the exterior.

D Fixed-cutoff differentiation, pole counting, and coarea

This appendix gives the differentiation rule used in the intermediate sectors. The integration domain and real cutoffs stay fixed. A transport field moves the parameter derivative away from the two global phases, leaving simple kernels. The recurrence then counts the remaining poles, and the flux estimates justify every integration by parts before coarea.

D.1 The identity on a fixed real integration domain

Lemma D.1 (Fixed-domain differentiation)**.** *Let w be a smooth, s-independent weight on a real angular torus. For an s-dependent complex vector field V and a top form A_s, set $\mathcal{D} = \partial_s - \mathcal{L}_V$. If all puncture fluxes vanish at every intermediate stage through order j, then*

$$\partial_s^j \int wA_s = \int \mathcal{D}^j(wA_s).$$

If $(\partial_s - V\cdot\nabla)Y = (\partial_s - V\cdot\nabla)Z = 0$, the factors $(1-Y)^{-1}$ and $(1-Z)^{-1}$ remain simple throughout this identity.

Proof. Apply the transport identity once on a punctured fixed domain, retain its boundary flux, and then iterate. The assumed flux limits are required at every stage, which is why the sector arguments verify them through the full derivative order.

On a top form, $\mathcal{L}_V = \mathrm{d}\iota_V$ is a divergence. Its integral is the boundary flux, including every puncture boundary. Its vanishing proves the first derivative identity. Apply that identity to the entire s-dependent top form produced at the preceding step. In particular, the iteration differentiates V, its divergence, and all coefficients already present. The kernel assertion follows from the derivation rule. Complex coefficients in V are allowed by linearity of real integration by parts. No real flow or holomorphic extension of w is required. □

If $\sum_h w_h = 1$ is a fixed smooth partition, use a different field V_h on each weighted term:

$$\partial_s^j \int A_s = \sum_h \int (\partial_s - \mathcal{L}_{V_h})^j (w_h A_s). \tag{D.1}$$

No equality between the fields on overlapping supports is required. The proof is the preceding lemma applied to each integral, followed by addition. This observation avoids interpolating fields in a way that could destroy either frozen-product identity.

D.2 Regular branch coordinates and the selected-pair field

On the original all-branch chart let $\chi_s : u \mapsto \phi$, where $\phi = \arccos W$. The inverse $y_s(\phi)$ is even and analytic on a fixed polydisk about $(s_*, 0)$, because $W_\theta \neq 0$ there. The one-body form and the complete pair are

$$\frac{\mathrm{d}y}{2\pi\mathrm{i}}\frac{z}{\sinh\gamma} = -\frac{z}{\pi(1-y^{-2})}\,\mathrm{d}\phi, \qquad P_{il} = (\phi_i - \phi_l)^2 A_s(\phi_i, \phi_l). \tag{D.2}$$

Here A_s is analytic with bounded fixed jets. Indeed, $y_s(\phi_i) - y_s(\phi_l)$ has the two factors $\phi_i - \phi_l$ and $\phi_i + \phi_l$; the latter cancels the corresponding factor in $1 - e^{-\mathrm{i}(\phi_i+\phi_l)}$. No division by a remaining collision zero is used.

Write the regular top form as

$$\Omega_s = \frac{\mathscr{N}_s(\phi)}{\mathcal{K}_s(\phi)}\,\mathrm{d}\phi_1 \cdots \mathrm{d}\phi_N, \qquad \mathcal{K}_s = (1-Y_s)(1-Z), \qquad \mathscr{N}_s = \Delta(\phi)^2 \mathscr{A}_s(\phi).$$

The analytic factor $\mathscr{A}_s$ includes all normalized measures, the inverse-product numerator, and the regular pair coefficients.

Put $a_i = S'/W_{u_i}$, $b_i = \partial_{u_i}\phi_i$, and $A = \sum_i a_i$. For a selected pair p, q, set

$$\begin{aligned} V_i &= a_i \quad (i \neq p, q), \\ V_p &= a_p + \frac{Ab_q}{b_p - b_q}, \qquad V_q = a_q - \frac{Ab_p}{b_p - b_q}. \end{aligned} \tag{D.3}$$

Then $\sum_i V_i = 0$ and $\sum_i b_i V_i = \partial_s \sum_i \phi_i$, so Y, Z are frozen. In regular coordinates the residual field $B = \partial_s \chi_s - (\chi_s)_* V$ has only two nonzero entries:

$$B_p = -\frac{Ab_p b_q}{b_p - b_q}, \qquad B_q = -B_p.$$

Since $b(\phi) = B_s(\phi)/\phi$ with B_s even, analytic, and nonzero near zero,

$$\frac{b_p b_q}{b_p - b_q} = \frac{H_s(\phi_p, \phi_q)}{\phi_q - \phi_p}.$$

The analytic coefficients and any fixed number of their derivatives have polynomial N-bounds. Both V, expressed in regular coordinates, and B have at most one selected-difference pole; $\mathrm{div}_\phi B$ has at most two.

The pullback identity needed for smooth real cutoffs is

$$(\partial_s - \mathcal{L}_V)(w\chi_s^*\Omega_s) = w\chi_s^*(\partial_s + \mathcal{L}_B)\Omega_s - (V \cdot \nabla_u w)\chi_s^*\Omega_s. \tag{D.4}$$

Writing $J_s = \det D_u \chi_s$, we have $\partial_s J_s = J_s(\mathrm{div}_\phi(\partial_s \chi_s \circ \chi_s^{-1})) \circ \chi_s$. The corresponding density formula for $\mathcal{L}_V$ is $V \cdot \nabla_u + \mathrm{div}_u V$. Subtracting these two product rules yields (D.4), including the displayed derivative of the real weight. In particular, $(\partial_s + B \cdot \nabla_\phi)\mathcal{K}_s = 0$. Use (D.4) on the real u-domain, where w is a smooth angular weight.

D.3 The complete high-order recurrence

Lemma D.2 (Unfactored jet filtration)**.** *For each fixed j, the j-fold operation in* (D.4) *is a finite sum of terms*

$$(\partial_u^\mu w)(u)\,\chi_s^*\left[\frac{c_{j,\mu,\ell,\nu}(s,\phi)\,\partial_s^\ell \partial_\phi^\nu \mathscr{N}_s(\phi)}{\mathcal{K}_s(\phi)}\,\mathrm{d}\phi_1 \cdots \mathrm{d}\phi_N\right]. \tag{D.5}$$

If m is the selected-difference pole order of a displayed coefficient, then that term can be chosen to satisfy

$$m + |\mu| + |\nu| \leq 2j, \qquad \ell + |\mu| + |\nu| \leq j. \tag{D.6}$$

Apart from these explicitly counted poles, fixed coefficient jets are bounded by $C_j N^{C_j}$ on the fixed regular-coordinate neighborhood. There are at most $C_j N^{C_j}$ terms.

Proof. Expand one application of the full transport operator and track how it changes the numerator derivatives and the pole powers. The five cases below preserve the two displayed filtration inequalities at every iteration.

For $j = 0$ take $c = 1$. Suppose (D.5) holds. Apply (D.4) to each term, and use the fact that the kernel is frozen. The possibilities are as follows:

1. A fixed-ϕ s-derivative on c does not raise its difference-pole order: $\delta = \phi_p - \phi_q$ is independent of s in these coordinates.
2. $B \cdot \nabla_\phi c$, or multiplication by $\operatorname{div}_\phi B$, raises m by at most two.
3. $B \cdot \nabla_\phi$ on the numerator derivative raises $|\nu|$ by one and m by at most one.
4. ∂_s on the numerator derivative raises ℓ by one without adding a difference pole.
5. The final term of (D.4) raises $|\mu|$ by one and contributes one component of V, whose pole order is at most one.

Every operation therefore raises the first count by at most two and the second count by at most one. These are all product-rule terms, including derivatives of the previous field coefficients. Bounded analytic coefficients, at most N summands in A, and a fixed number j of differentiations give the claimed polynomial term and coefficient bounds. □

The object differentiated in this recurrence is the *unfactored numerator* $\mathscr{N}_s$. A numerator derivative of order $\ell + |\nu| \le j$ changes at most j pair factors. The remaining pair contractions survive. Fixed-ϕ s-derivatives preserve the Vandermonde polynomial; $|\nu|$ coordinate derivatives reduce its full-collision degree by at most $|\nu|$. The recurrence retains the differentiated numerator. Dividing by the original Vandermonde would introduce poles at unselected collisions.

D.4 Rational weights and all boundary fluxes

Fix an integer $M > 2k + 2$, independent of N, and put

$$d = \max_{i<l} |u_i - u_l|, \qquad w_{pq} = \frac{(u_p - u_q)^{2M}}{\sum_{i<l}(u_i - u_l)^{2M}} \quad (d > 0).$$

They sum to one off full equality. For $|\mu| \le k$,

$$|\partial_u^\mu w_{pq}| \le C_k N^{C_k} \frac{|u_p - u_q|^{2M-|\mu|}}{d^{2M}}. \tag{D.7}$$

To see this, differentiate the quotient. A derivative of the denominator contributes a power at most d^{-1}; replace it, when needed, by $|u_p - u_q|^{-1}$. The denominator itself is at least d^{2M}, and its derivative sums have polynomial N-cost. This proves (D.7) term by term.

On the original branch cone, $|\phi_p - \phi_q| \ge c|u_p - u_q|$, by Lemma 7.4. Combining (D.7) with (D.6) absorbs every selected pole at a cost $C_j N^{C_j} d^{-2j}$ on a separated support, while the undifferentiated pair factors retain their quadratic contraction. Other fixed smooth cutoffs are included by the ordinary product rule. A cutoff at scale b_N or ρ_N has its corresponding inverse-scale cost; these costs are retained in Propositions 7.5 and 7.6.

Near full equality, the numerator has degree $N(N-1)$. The bound $m + |\mu| + |\nu| \le 2j$ shows that a full j-th density loses at most $2j$ powers of the shape radius. A further inverse radius is

reserved when a boundary flux or a transition-cutoff estimate requires it. This is the origin of the conservative loss $2j+1$ used in the sector proofs.

First take a selected-diagonal tube of radius h on a region where d is bounded below. The pair double zero, the $2M$-power weight, the preceding j operations, and one extra field factor give a flux at most

$$C_{\varepsilon,N,j}\, h^{2M+1-2j}.$$

It vanishes. Next take a full-equality sphere of radius ρ. At fixed $\varepsilon > 0$ all global denominators and regular coordinate changes are bounded; the shape surface measure contributes ρ^{N-2}. A sufficient flux exponent is

$$N(N-1) - 2j - 1 + N - 2 = N^2 - 2j - 3 > 0 \quad (N \geq N_0 + 2,\ j \leq k).$$

The limits are taken in this order at fixed ε, N. There are no boundaries at a cutoff support endpoint because the cutoffs are smooth on the closed torus. These flux checks apply at every earlier stage $0 \leq l \leq j$, and establish the iterated identity before any boundary-uniform estimate is taken.

D.5 Mixed and compact charts

In a mixed chart only the true branch coordinates are replaced by ϕ_i. The selected denominator is $1 - b_q/b_j$, bounded away from zero on the nested support, and $1/b_j$ is analytic at the branch. There is consequently no selected-collision pole in this calculation. The same pullback recurrence includes the regular branch one-body form, the compact-angle measures, and the full contour Jacobian. All fixed jets have polynomial N-cost, apart from C^N one-body products.

For the compact right chart the coefficients are smooth in real angles and analytic in s. The factorization $D_{pq} = (\theta_p - \theta_q)C_{pq}$, with $|C_{pq}| \geq c > 0$, uses smooth division, as proved in Proposition 8.2. The same five product-rule cases prove a two-power loss per operation in the real angular variables, with the deformation bumps differentiated there as smooth functions.

Coupled occupancy and the number of changed pairs. The compact-right current field may move a coordinate where $p' \neq 0$. The exact contour Jacobian gives

$$(V \cdot \nabla \log y_i)_{\mathrm{occ}} = \frac{\lambda\tau}{2N} m_i \sum_l V_l p'_l.$$

An individual off-diagonal entry is $O(\lambda\tau/N)$. The compact-right field has two nonzero components, each possibly $O(N/|\theta_p - \theta_q|)$, so the contraction can be $O(\lambda\tau/|\theta_p - \theta_q|)$. It moves every m-supported root and must be retained. All these roots are compact in this chart; no singular raw branch-angle jet is used.

To count descendants, keep each complete pair $F_e = P_{il}$ as a scalar function of all angles and s, including its occupancy dependence. Write the density as $G\prod_e F_e$. The scalar operator $\partial_s - V\cdot\nabla$ is a derivation; the top-form correction is multiplication by $-\operatorname{div} V$. In one product-rule summand a step hits a coefficient or G, an already changed pair, or one previously unchanged pair. Divergence changes none. Induction therefore leaves at most j changed pairs per descendant at order j, even though the sum of descendants involves all pairs. In regular coordinates, hitting either the double-zero polynomial or its analytic coefficient marks that same pair. No division by an original pair is involved.

There are $O(N^2)$ possible pair choices at a step. Expanding a coordinate contraction or an occupancy sum introduces at most $O(N)$ further choices. A fixed number of chain rules therefore has $N^{O(j)}$ descendants. Fixed jets of P/N are bounded by the fixed bump jets, and fixed compact root and pair jets are bounded on the continuation neighborhoods of Lemma 7.2. Sums of phase jets, and inverses of the separated smooth divided determinant C_{pq}, have only polynomial fixed-jet costs. The explicit selected difference pole still adds at most two powers per Lie step. One-body and Jacobian products cost $C_j^N N^{O(j)}$. Consequently the coupled terms fit the existing polynomial coefficient and term budgets and the unchanged pairs retain $C_j^N e^{-\kappa N^2}$. In mixed Stokes charts the plateau identities $V \cdot \nabla P = V \cdot \nabla \sum m_i = 0$ remove this coupling; original K has $\lambda = 0$.

D.6 Simple kernels and the order of coarea

For $a > 0$,

$$\frac{1}{|1 - e^{-a+it}|} \le \frac{C}{\min(1, a) + \operatorname{dist}(t, 2\pi\mathbb{Z})} \quad (0 \le \operatorname{dist}(t, 2\pi\mathbb{Z}) \le \pi).$$

On an interval of length L, the integral is at most $C(1 + L)(1 + |\log a|)$ for $0 < a < 1$. A variable damping can be replaced by its positive lower bound in this estimate.

For two unwrapped phases F, G, suppose their image is contained in $I_F \times I_G$, the map has multiplicity at most m_0, and its real Jacobian has already been divided out in the density bound. If the attenuation floors are $a_Y, a_Z \in (0, 1)$, the remaining two-kernel integral is bounded by

$$C m_0 (1 + |I_F|)(1 + |I_G|)(1 + |\log a_Y|)(1 + |\log a_Z|).$$

Indeed extend the nonnegative integrand to the containing rectangle and apply the one-dimensional bound twice. Equivalently, subdivide each interval into translates of one period. This accounts for periodic wrapping separately from the multiplicity of the unwrapped map. In the near-equality sectors $m_0 = 1$ on each fixed shape ray and both lengths are at most CN, yielding an explicit CN^2 factor. Any already counted index cells or fixed-jet terms retain their stated polynomial costs. Such factors are absorbed by the existing $e^{C_j N \log(N+1)}$ budgets.

Apply coarea to the final absolute integral, using the real Jacobian and multiplicity specified in each sector. Introduce the sign and extreme-index cells at that stage. When the phase replaces the shape radius, bound the remaining positive power of the radius on the support before integrating the phase.

E Constant and quantifier dependencies

Fix the constants in the following order to retain the support and pair margins. Each step uses quantities selected earlier and precedes the final restriction on ε. The choices are independent of N.

1. Fix (p, a, b), hence N_0, k, S_*, θ_B, and the original radial constant and a preliminary bound on $|S'|$. The final disk constants are chosen in step 4. There are only finitely many derivative orders $0 \le j \le k$.

2. Fix a sufficiently small branch neighborhood and compact margins. Choose the nested ratio L large enough for the outer-anchor/true-branch slope separation, also relative to the

bounded upper-current slopes on a preliminary endpoint neighborhood separated from the upper branch. Fix the smooth nested partition and all of its finite derivative bounds.

3. Obtain the same-group pair constant $q < 1$ at the limiting real support, with the groups and continuation neighborhoods of Lemma 7.2. Choose $q < q_1 < q' < 1$, then the cross-group slack η, and then the upper endpoint slivers and deformation strength τ, retaining the reserved same/cross disk slack. The upper branch must lie inside $a = p = 1$. Fix the bump jets and the positive constants b_τ, c_q at this step.

4. Choose the selected c/N, original $c\varepsilon$, and protected-current $c\lambda_*/N^2$ disk constants small enough for the already fixed margins. In particular impose $L_S c_O \le b_0/4$, $L_S c_P \le b_\tau/4$, the compact-distance and reserved pair-slack restrictions, and $C c_P \le c_q/2$ from Lemma 8.3. No reduction of τ follows this choice. Take a common smaller positive κ for the finitely many pair estimates.

5. Choose C_H from the original Pfaffian bound, $0 < \alpha_0 < 1/[2(k+2)]$, and then D with $\kappa D^2 > k + 2$. Fix the rational-weight integer $M > 2k + 2$ and a sufficiently large compact-equality exponent $K_0 > 1$.

6. Choose the resultant exponent A, followed by the small microcore prefactor c_* in $b_N = c_* e^{-(A+4)N}/N$. Determine the finite-jet constants on the anchor neighborhood $\rho \le b_N/8$, $|u_a| \ge b_N/2$. Its lower bound on $|t|$ determines the cubic coefficient C independently of the still unselected exponent B.

7. With the finite Γ_* defined in Proposition 7.6, choose $B, C_1 > A + 4 + \max\{1, \Gamma_*\}$. This ensures the geometric inclusions $2\rho_N \le b_N/8$ and $\delta_N \le b_N/16$ for every $N \ge N_0 + 2$, including the finitely many low orders. Increase B further so that $\exp((A+4)N^3/2 + \frac{1}{2}N^2 \log N + C_j N^2)(2e^{-BN})^{N^2-2j-4}$ is controlled for all large N and every $j \le k$. Only the remaining numerical bounds at finitely many orders are absorbed into a point-dependent constant; their support geometry already satisfies the inclusions.

8. Finally decrease a single $\varepsilon_0 > 0$ for all the conditions. In the intermediate window, for every fixed C, m, r,

$$\varepsilon^r e^{CN} N^m \longrightarrow 0 \quad (r > 0), \qquad \varepsilon/b_N^m \longrightarrow 0, \quad N \le D\sqrt{\log(1/\varepsilon)}.$$

   These limits also give $N^{C_j}(\varepsilon + \lambda_*) \to 0$. They are used only in that window. In particular impose $N\varepsilon e^{AN} \le c$, $\varepsilon/b_N \le c$, and the finitely many compact-jet perturbation inequalities for $j \le k$, as well as $\varepsilon + \lambda_*$ sufficiently small relative to the fixed branch and compact margins. For the first two the logarithms at $N \le D\sqrt{H}$ are bounded by $-H + O(\sqrt{H} + \log H)$; for the compact perturbations the bound is $-\alpha_0 H + O(\log H)$. Thus one ε_0, independent of N, enforces every required condition. Only finitely many fixed exponents from the preceding estimates are needed.

The ordinary one-body products cost C_j^N, and the full collision coefficients may cost $e^{C_j N^2}$. The conversion from the exponentially small branch anchor to shape coordinates may cost e^{CN^3}. The positive collision power at ρ_N absorbs this last factor.

Quantity	May depend on	Uniformity
c_0, c_1, original radius/disk	Fixed selected point	All original angles and all even N; radius held fixed under each s-derivative.
Branch and compact margins	S_*, θ_B, k	Independent of N, ε; compact groups exclude the exact branch before strict constants are chosen.
q, η, κ, τ	The preceding margins and point	Entire stated support, all occupancies $P/N \in [0,1]$; radial centers and both stated disks.
Selected-contour disk c/N	Point, margins, τ	All sufficiently small ε on the selected deformed contour.
Current disk $c\lambda_*/N^2$	Point, margins, τ	$\lambda_* \le \lambda \le 1$; compact roots may leave the unit disk.
C_j in jets	Point, fixed bumps, $j \le k$	Independent of N, ε within the stated window; powers N^{C_j} are explicit.
Final C_j in (5.7)	Point, j, all preceding margins, $\tau, \kappa, \alpha_0, D, M, K_0, A, c_*, B, C_1$	Independent of N, ε, λ. Finite small-order constants may be large but are ε-independent.
A, b_N, ρ_N	Algebraic point; B also on k	Exponential N-dependence is retained. $\varepsilon e^{cN} \to 0$ only for $N \le D\sqrt{H}$.
α_0, D, C_H	Point and k	Fixed before $\varepsilon \to 0$; all lower derivatives use the same finite choices.
First-chart constants	The fixed N_0, p, a, b	Fixed selected chart.

Every estimate stated for sufficiently small ε means

$$\forall (p,a,b)\ \exists \varepsilon_0, C_j, \dots\ \forall\, 0 < \varepsilon < \varepsilon_0\ \forall\, N \text{ in the specified window.}$$

Split the differentiated integrals into the N- and λ-windows at their stated thresholds.

E.1 Tail-sector decomposition

In the table, intermediate means $N_0 + 2 \le N < D\sqrt{H}$; high means $D\sqrt{H} \le N < C_H(H+1)^2$. All rows apply to every $0 \le j \le k$. The N- and λ-splits are made after differentiation.

Sector	Region	Support and estimate	Proof location
1	Ultra-high original	Every angle, $N \ge C_H(H+1)^2$; weighted matched pairs and original disk.	Proposition 5.2
2	Final selected F	Named $a_q > 0, p_q = 1$; protected branches; compact matching penalty; disk c/N.	Proposition 6.1

Sector	Region	Support and estimate	Proof location
3	High K and small-current orders	Enlarged lower support, with one named upper transition variable for the current; original $c\varepsilon$ disk.	Lemma 7.3, Proposition 8.5
4	Intermediate all-branch core	Every $\lvert u_i\rvert \le b_N$; resultant Y-gap and one-phase branch transport.	Proposition 7.5
5	All-branch far complement	Named $\lvert u_a\rvert \ge b_N/2$, $\rho \ge \rho_N$; rational weights and extreme-pair coarea after differentiation.	Proposition 7.6
6	All-branch near equality	Same anchor, $\rho \le 2\rho_N \ll b_N$; common sign and positive full-collision exponent.	Proposition 7.6
7	Intermediate K, compact left	Outer anchor away from branch and a compact left root; fixed Z-gap and one simple phase.	Proposition 8.1
8	Intermediate K, mixed	Outer compact q, true-branch j, $\lvert b_j\rvert \ge 4\lvert b_q\rvert$; two-phase field at $\lambda = 0$.	Proposition 8.1
9	Intermediate K, all right	No true branch or left label; strict real phase curvature and compact rational atlas.	Proposition 8.2
10	Intermediate small current, left	Named q on an upper plateau and a left modulus gap; all moving branches remain on true plateaus.	Proposition 8.1
11	Intermediate small current, mixed	Upper-right q, true-branch j; exact preservation of both occupancy sums in every derivative.	Proposition 8.1
12	Intermediate small current, all right	Only compact right variables; coupled Hessian perturbation $N^{C_j}(\varepsilon + \lambda_*) \to 0$.	Proposition 8.2
13	Large current	Named q, $P \ge 1$, $\lambda \ge \lambda_*$; disk $c\lambda_*/N^2$, root inflation and product gaps.	Lemma 8.3, Proposition 8.4

The upper inverse branch $+\theta_B$ is in the $a = p = 1$ plateau. It can appear in F, where it is shifted inward; it cannot appear on a K or Stokes transition support. The nested partition in Lemma 7.1 keeps every derivative of a compact label out of the inner branch region.

F Normalization and the independent angular representation

This appendix connects the offsite contour coefficients to the full physical observable. It keeps track of the onsite term, the four angular charts, and both physical prefactors. The amplitude comparison uses the coefficient derived in Section 4 on the same physical branch. The squared formula in Theorem 4.1 of [8] reduces on p. 375 to $\det_2(I+G) = \det_2(I+g)^2$. For the smooth trace-class two-point kernel, the zero diagonal gives $\operatorname{Tr} g = 0$, hence $\det_2(I+g) = \det(I+g)$. The determinant is real and continuous for real $s > 1$; the Griffiths bound used on p. 359 keeps its modulus at least one, and its limit one at infinity fixes the positive square root. The resulting unsquared series is Theorem 5.0 and its proof on pp. 374–378. In the representation of the published Appendix 1 of [10], it has N factors $\mathrm{d}\theta/(2\pi)$. With $x = e^{\mathrm{i}\theta}$, these become $\mathrm{d}x/(2\pi\mathrm{i}x)$. Introducing N auxiliary residues uses another N factors $\mathrm{d}y/(2\pi\mathrm{i})$. Thus the double contour prefactor is $(2\pi\mathrm{i})^{-2N}/N!$ for ordinary unnormalized contours, as displayed with the published

equation (3) of [10]. Summing the nonorigin lattice sites gives

$$\frac{4}{(1-X)(1-Y)} - \frac{2}{1-X} - \frac{2}{1-Y} = \frac{2(X+Y)}{(1-X)(1-Y)}.$$

The factors $1/(XY)$ in the normalized correlation measures then give the numerator in (2.1) and the factor two in (2.2); its physical origin term is $\langle \sigma_{00}^2 \rangle - \mathcal{M}^2 = 1 - \mathcal{M}^2$.

For fixed real $s > 1$, choose a small common angle strip with $\Re\gamma \geq c > 0$ and bounded canceled kernel. Shifting all angles by $i\eta\,\mathrm{sgn}(a)$ and applying Hadamard bounds the N-particle coefficient at the lattice site (a, b) by

$$\frac{N^{N/2} C^N}{N!} e^{-N(\eta|a|+c|b|)}.$$

The determinant integrand is periodic, and the bound is summable in sites and even $N \geq 2$. This justifies the lattice/Fredholm interchange on the real interval; it supplies no boundary-uniform infinite-tail estimate.

For fixed even N, define C_N^{std} as the full-site coefficient $\chi^{(N)}$ displayed with the published equation (3) of [10]. Define $f_{00}^{(N)}$ as its onsite Fredholm coefficient: the integral in (2.1) with its global rational factor replaced by $1/(XY)$, retaining the same pair factors, factorial and measures. The exact numerator identity

$$\frac{(1+X^{-1})(1+Y^{-1})}{(1-X)(1-Y)} = \frac{1}{XY} + \frac{2(X^{-1}+Y^{-1})}{(1-X)(1-Y)}$$

therefore gives, without summing over particle number,

$$C_N^{\mathrm{std}} = f_{00}^{(N)} + 2T_N.$$

The local convex-gradient argument with both global product factors omitted bounds every fixed derivative of the onsite term at the selected point.

The same observable in the original literature. The low-temperature expansion (45) of [4] places $\mathcal{M}^2$ outside the sum of $\chi^{(N)}$. Its cited integral source, [7, (1.10), (3.13), (3.17)–(3.20)], has the full angular torus, the factorial $1/N!$, and the factor $(1+Z)/(1-Z)$. At isotropy its anisotropy factor $\cot\alpha$ is one; that source's anisotropy symbol is unrelated to our geometric angle. Thus its $\widehat{\chi}^{(N)}$, the review's $\chi^{(N)}$, and our C_N^{std} are the same normalized full-site coefficient. Equations (4)–(9) of [2] independently give this form:

$$C_N^{\mathrm{std}} = \frac{1}{N!} \int \prod_{i<N} \frac{\mathrm{d}\varphi_i}{2\pi} \prod_i \frac{1}{\sinh\gamma_i} \frac{1+Z}{1-Z} \prod_{i<j} \left[\frac{\sin((\varphi_i - \varphi_j)/2)}{\sinh((\gamma_i+\gamma_j)/2)} \right]^2.$$

Here $\sum_i \varphi_i = 0 \pmod{2\pi}$, $\cosh\gamma_i = S - \cos\varphi_i$, and $z_i = e^{-\gamma_i}$. The last source's $w = s/[2(1+s^2)]$ satisfies $1/(2w) = S$. Its x_i is our z_i, and its $y_i = 1/\sinh\gamma_i$ includes the numerator $2w$.

The physical branch and four angular charts. Continue the positive root $\gamma_i > 0$ from real $s > 1$. For $s = re^{i\theta_*}$, $r > 1$, $\Im S = (r - r^{-1})\sin\theta_* > 0$. The root

$$z = (W + \sqrt{W-1}\sqrt{W+1})^{-1}, \qquad W = S - \cos\varphi,$$

with principal square roots is the analytic interior root in this upper half-plane. No root can cross the unit circle there, since $z + z^{-1} = 2W$; continuation from large positive W fixes the interior root. The same equation then forces $\Im z < 0$. It has $|z| < 1$ and $\Im z < 0$, and tends to $e^{-\mathrm{i}\beta}$ at $\varphi = \pm\alpha$. On this continuation $\gamma \to \mathrm{i}\beta$. Writing $\gamma = \mathrm{i}u$, implicit differentiation gives, at $\varphi = \alpha$,

$$u_\varphi = -\frac{\sin\alpha}{\sin\beta}, \qquad u_{\varphi\varphi} = -\frac{H_{\alpha\beta}}{\sin^3\beta}, \qquad s_* u_s = -\frac{2\mathrm{i}\sin\theta_*}{\sin\beta}.$$

For $\varphi_i = \alpha + \tau_i$, $\tau_N = -\sum_{i<N}\tau_i$, the linear angular sum vanishes. With $a_0 = (N^2-1)/2 = k+1/2$, $\nu = k - 1/2$, $Q = 2N\sin\theta_*/\sin\beta$, and $d = H_{\alpha\beta}/(2\sin^3\beta) > 0$, the denominator begins as

$$1 - Z = Q\varepsilon - \mathrm{i}d\sum_i \tau_i^2 + O(\varepsilon^2 + \varepsilon|\tau| + |\tau|^3).$$

Each one-body coefficient is $-\mathrm{i}/\sin\beta$, and each squared pair contributes $-1/(4\sin^2\beta)$. For $N = 2m$ their total phase is $(-\mathrm{i})^N(-1)^{N(N-1)/2} = (-1)^{2m^2} = 1$. The numerator $1 + Z$ tends to two. Thus the chart prefactor is

$$p_0 = \frac{2\,2^{-N(N-1)}}{N!(2\pi)^{N-1}\sin^{N^2}\beta} = 2\pi K_\beta.$$

This is the same constrained coordinate measure and normal form as after our mean residue. Its k-th complex derivative therefore has coefficient L_β, with the finite-jet and remainder control already used in Section 4. The factor $\sqrt{N}$ for converting hyperplane area to coordinate measure is already included in $\mathscr{C}_N$.

The selected point has exactly the two distinct nonzero cosine values from Theorem 3.1. In this angular representation they give four signed centers $+\alpha, -\alpha, +\beta, -\beta$. All satisfy the periodic constraint because $N\alpha, N\beta \in 2\pi\mathbb{Z}$. Angle reversal preserves the real measure and squared Vandermonde; the α/β exchange preserves (B.1). Thus all four coefficients agree. Particle permutations fix each all-equal center, so no further $N!$ is attached to this count. Completeness is also forced by the already proved identity $C_N^{\mathrm{std}} = f_{00}^{(N)} + 2T_N$: the two offsite charts give $2L_\beta$, and the onsite term has bounded fixed derivatives. There is no unestimated same-order contribution in this comparison.

Proposition F.1 (Complete-amplitude conversion)**.** *Fix the selected prime family and its first even order $N = 2p$. In the review's index convention take $j = 2a$, $l = 2b$, so that $\phi^{(j)} = \alpha$, $\phi^{(l)} = \beta$. For $\vartheta = \pm\theta_*$, define the expressly evaluated printed scalar*

$$R_N(\vartheta) = \frac{(N\mathrm{i}\sin\vartheta)^\nu P_N}{H_{\alpha\beta}^{a_0}\pi^{(N-3)/2}\sqrt{N}\,\Gamma(a_0)},$$

using the principal power in the numerator and the positive real power of $H_{\alpha\beta}$. This is (69) in the arXiv version of [4], or (70) on p. 801 of the journal version; its angle relation is respectively (70) or (71). The root is fixed by $S_* = \cos\alpha + \cos\beta$, as in [7, (3.24)]. This root agrees with the explicit root and cosine relation following equation (64) of the review, where the printed plus sign is inconsistent with those formulas. On the physical low-temperature exterior germ put $\delta = 1 - s_*/s = \varepsilon/(1+\varepsilon) > 0$. Modulo a radial Taylor polynomial of degree at most $k-1$, the

coefficient of δ^ν is

observable	*coefficient*
one chart	$A_{\mathrm{chart}} = R_N(-\theta_*) = -\mathrm{i}R_N(\theta_*)$
T_N	$2A_{\mathrm{chart}}$
$C_N^{\mathrm{std}} = \chi^{(N)}$ *of the review*	$4A_{\mathrm{chart}}$
physical normalized bulk $\mathcal{X}$	$4\mathcal{M}^2(s_*)A_{\mathrm{chart}}$
physical χ	$4\beta_{\mathrm{phys}}(s_*)\mathcal{M}^2(s_*)A_{\mathrm{chart}}$

The last two rows use the physical identification and the first-order and tail results of Theorem 9.2. Before multiplication by $\mathcal{M}^2$, the full offsite lattice coefficient is $2T_N$, whereas T_N denotes the lower double-contour coefficient. Its full offsite singular coefficient equals that of C_N^{std}. The corresponding k-th complex derivative coefficients are $L_\beta, 2L_\beta, 4L_\beta, 4\mathcal{M}^2(s_*)L_\beta$, and $4\beta_{\mathrm{phys}}(s_*)\mathcal{M}^2(s_*)L_\beta$.

Proof. Taking k radial derivatives multiplies a complex derivative by s_*^k. If $f^{(k)}(\varepsilon) = B\varepsilon^{-1/2} + o(\varepsilon^{-1/2})$, its integrability gives finite limits for all lower derivatives. The integral Taylor remainder is

$$\frac{1}{(k-1)!}\int_0^\varepsilon (\varepsilon - t)^{k-1} f^{(k)}(t)\,\mathrm{d}t = \frac{B\sqrt{\pi}}{\Gamma(k+1/2)}\varepsilon^{k-1/2} + o(\varepsilon^{k-1/2}).$$

The same estimate with the error bounded by $\sup_{0<t<\varepsilon} o(1)\,t^{-1/2}$ justifies the last little-o. Changing to $\delta = \varepsilon/(1+\varepsilon)$ changes the Taylor polynomial but preserves this first fractional coefficient. In particular

$$A_{\mathrm{chart}} = s_*^k L_\beta \frac{\sqrt{\pi}}{\Gamma(a_0)}.$$

Equation (B.1) gives its positive magnitude factor and phase $(-1)^k e^{\mathrm{i}\pi a_0/2}$. For even N, $k = 2(N/2)^2 - 1$ is odd, hence this phase equals $e^{-\mathrm{i}\pi\nu/2}$. The latter is exactly the principal power phase at $\vartheta = -\theta_*$; at $+\theta_*$ the ratio is $-\mathrm{i}$. The geometric denominator and every factorial factor are identical, proving the scalar identities. The two/four chart counts, bounded onsite derivative and bulk product rule above give the remaining rows. The Taylor integration also applies to the full bulk because its already proved k-th derivative has precisely the indicated leading term.

The phase can also be read from the radial normal-form integrand $r^{2k}/(q + br^2)$, with $q = Q\varepsilon$ and $b = -\mathrm{i}d$. Polynomial division leaves

$$\int_0^\eta \frac{r^{2k}}{q+br^2}\,\mathrm{d}r = \text{polynomial in } q + \frac{(-q)^k}{b^k\sqrt{qb}}\arctan\!\left(\eta\sqrt{b/q}\right).$$

Continue b from the positive real axis to $-\mathrm{i}d$, with $q > 0$. The last factor tends to $\pi/2$; its noninteger term is $(\pi/2)(-1)^k q^{k-1/2} b^{-k-1/2}$. Multiplying by $p_0\mathscr{C}_N$ gives the same coefficient. Section 4 controls the error in replacing the original integral by this normal form. □

The physical prefactors have fixed branches too. On this quadrant, the continuation with $\mathcal{M}^2 \to 1$ at infinity has

$$\mathcal{M}^2(s_*) = (2\sin 2\theta_*)^{1/4} e^{\mathrm{i}(\pi/8 - \theta_*/2)}, \qquad \beta_{\mathrm{phys}}(s_*) = \frac{\operatorname{arsinh}(s_*)}{2J},$$

where arsinh is continued from positive real $s > 1$ through the right half-plane. These factors are analytic and nonzero locally. They are absent from the review's normalized $\chi^{(N)}$. For another

local coordinate $x = \lambda\delta + O(\delta^2)$, the coefficient instead acquires $\lambda^{-\nu}$ with the continued power branch; here $\lambda = 1$ for the two stated radial variables.

The printed scalar is the single-chart value for the selected distinct nonzero angles. The four chart contributions in the full-site $\chi^{(N)}$ coefficient have the same phase on the physical sheet, which is fixed by the integral continuation above. The resulting coefficient has four times the magnitude of the printed scalar for either angle sign in the review's half-integer powers. The comparison applies to these distinct nonzero angles on the stated physical sheet.

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